

Microprism-based two-photon imaging of the mouse inferior colliculus reveals novel organizational principles of the auditory midbrain

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Abstract

To navigate real-world listening conditions, the auditory system relies on the integration of multiple sources of information. However, to avoid inappropriate cross-talk between inputs, highly connected neural systems need to strike a balance between integration and segregation. Here, we develop a novel approach to examine how repeated neurochemical modules in the mouse inferior colliculus lateral cortex (LC) allow controlled integration of its multimodal inputs. The LC had been impossible to study via imaging because it is buried in a sulcus. Therefore we coupled two-photon microscopy with the use of a microprism to reveal the first-ever sagittal views of the LC to examine neuronal responses with respect to its neurochemical motifs under anesthetized and awake conditions. This approach revealed marked differences in the acoustic response properties of LC and neighboring nonlemniscal portions of the inferior colliculus. In addition, we observed that the module and matrix cellular motifs of the LC displayed distinct somatosensory and auditory responses. Specifically, neurons in modules demonstrated primarily offset responses to acoustic stimuli with enhancement in responses to bimodal stimuli, whereas matrix neurons showed onset response to acoustic stimuli and suppressed responses to bimodal stimulation. Thus, this new approach revealed that the repeated structural motifs of the LC permit functional integration of multimodal inputs while retaining distinct response properties.

eLife assessment

This study provides **important** new insights into how multisensory information is processed in the lateral cortex of the inferior colliculus, a poorly understood part of the auditory midbrain. By developing new imaging techniques that provide the first optical access to the lateral cortex in a living animal, the authors provide **convincing** in vivo evidence that this region contains separate subregions that can be distinguished by their sensory inputs and neurochemical profiles, as suggested by previous anatomical and in vitro studies. Additional information and analyses are needed, however, to allow readers to fully appreciate what was done, and the comparison of multisensory interactions between awake and anesthetized mice would benefit from being explored in more detail.

Introduction

Real-world listening occurs in environments cluttered by noise. To navigate these conditions, the auditory system relies on the integration of multiple sources of input. Broad integration of neural inputs, however, comes at a potential cost. Dense convergence can lead to inappropriate cross-talk between input streams, generating a processing “soup” whereby individual neural pathways cannot be independently modulated. Therefore, highly connected neural systems need to preserve specificity of inputs in the face of massive integration (Fair et al., 2007 [↗](#); Sporns, Tononi, & Edelman, 2000 [↗](#); Tononi, Sporns, & Edelman, 1994 [↗](#))

Several brain systems have solved the integration/segregation problem by creating repeated structural motifs that allow independent modulation of different pathways. For example, in the basal ganglia, neurochemical periodicity of the striosome and matrix regions permits retention of functional independence of input streams in the face of massive convergence. As a result, basal ganglia direct and indirect pathways may be independently modulated, while allowing controlled crosstalk between channels (Alexander, Crutcher, & DeLong, 1990 [↗](#); Brimblecombe & Cragg, 2017 [↗](#); Draganski et al., 2008 [↗](#); Gerfen, 1985 [↗](#))

In the auditory system, the integration/segregation challenge is particularly formidable in the inferior colliculus (IC). The IC serves as a specialized integration hub because of its heavy innervation by higher auditory centers as well as by nonauditory regions. Thus, the IC stands at the crossroads between multiple brain regions and is thus important for linking acoustic stimuli with non-acoustic inputs. The IC comprises a lemniscal division, the central nucleus (ICC), that receives primarily ascending auditory projections (Oliver, 2005 [↗](#); Willard & Ryugo, 1983 [↗](#); Willott, 2001 [↗](#)) and two non-lemniscal divisions known as the dorsal (DC) and lateral (LC) cortices that receive massive descending auditory as well as non-auditory input (Bajo & Moore, 2005 [↗](#); Coleman & Clerici, 1987 [↗](#); Loftus, Malmierca, Bishop, & Oliver, 2008 [↗](#); Saldaña, Feliciano, & Mugnaini, 1996 [↗](#); Schreiner & Winer, 2005 [↗](#); Winer, Larue, Diehl, & Hefti, 1998 [↗](#)) as reviewed in (A. M. H. Lesicko & Geffen, 2022 [↗](#)). The LC and DC appear to differ in their strategy for integrating these inputs. The DC does not contain known structural heterogeneities but has a tonotopic arrangement of neuronal responses on its surface (Barnstedt, Keating, Weissenberger, King, & Dahmen, 2015 [↗](#); A. B. Wong & Borst, 2019 [↗](#)). In contrast, the LC contains repeated neurochemical motifs, here referred to as modules, that express the inhibitory neurotransmitter GABA as well as other metabolic markers (Chernock, Larue, & Winer, 2004 [↗](#)). Areas outside the modules, referred to as matrix, stain strongly for calretinin (Dillingham, Gay, Behrooz, & Gabriele, 2017 [↗](#)). We and others have determined that nearly all of the long- and short-range connectivity of the LC is governed by the module/matrix organization (Dillingham et al., 2017 [↗](#); Lamb-

Echegaray, Noftz, Stinson, & Gabriele, 2019 [↗](#); A. M. H. Lesicko, Sons, & Llano, 2020 [↗](#); A. M. Lesicko, Hristova, Maigler, & Llano, 2016 [↗](#)). Unfortunately, how the LC integrates these inputs while maintaining distinct paths of information flow is not known. For example, LC neurons respond to multisensory stimuli (Jain & Shore, 2006 [↗](#); Zhou & Shore, 2006 [↗](#)), but information from different modalities enters the LC in segregated channels (A. M. Lesicko et al., 2016 [↗](#)). Thus, integration across the module/matrix border must occur, but the mechanisms of this integration are unknown. Traditional methods of studying the LC, such as in vivo electrophysiology, pose significant challenges in unambiguously assigning individual recorded neurons to residence in matrix or modules. Imaging approaches, such as two-photon (2P) microscopy, have proven to be enormously valuable in understanding the fine spatial structure of neural response properties in the auditory system (Bandyopadhyay, Shamma, & Kanold, 2010 [↗](#); Romero et al., 2020 [↗](#); Rothschild, Nelken, & Mizrahi, 2010 [↗](#)). However, because of its superficial location on the dorsal surface of the mouse midbrain, the DC is the only part of the IC that has been characterized by 2P imaging (Barnstedt et al., 2015 [↗](#); Ito, Hirose, Murase, & Ikeda, 2014 [↗](#); A. B. Wong & Borst, 2019 [↗](#)). Optically, it has been impossible to characterize the functions of the LC, which is laterally embedded deep in a brain sulcus. Here, we describe a novel approach using a 45° micropism inserted into the sulcus that separates the LC from the cerebellum to investigate the fine-scale functional organization of the LC from its lateral surface through 2P imaging. The 2P images obtained from the micropism revealed the first-ever images of the LC in vivo showing its characteristic GABAergic modules. Compared to the DC imaged from the surface, the acoustically and somatosensory-evoked calcium signals obtained by the 2P imaging of the LC through the micropism revealed functional distinctions between the different cellular motifs of the LC. For instance, while modules were more responsive to somatosensory stimulation than matrix, the latter was more acoustically-responsive and had a lower detection threshold for complex sounds. Additionally, modules and matrix were found to differently process the spectral and temporal features of the acoustic information. These findings suggest that the non-lemniscal divisions of the IC contain rich but distinct platforms for the processing of different features of auditory and non-auditory information.

Results

The insertion of 45° micropism for imaging the LC from the side

GAD67-GFP knock-in mice on a Swiss Webster background were crossed with Thy1-jRGECO1a mice to produce a GAD67-GFPx jRGECO1a hybrid mouse, which was used to visualize GABAergic modules, to identify the dense network of GABAergic neurons and terminals in modules of the LC in the green channel (Tamamaki et al., 2003 [↗](#)), and the calcium indicator, jRGECO1a, in the red channel (Dana et al., 2018 [↗](#)). A standard craniotomy was performed (Goldey et al., 2014 [↗](#)) with modifications to suit the specific characteristics of the IC in terms of its location, depth, and high density of the surrounding vasculature (Xiong et al., 2017 [↗](#)). The craniotomy (~ 3 x 4 mm) did not drill directly into the sagittal and lambdoid sutures which allowed minimal disruption of the superficial sinuses below the skull (**Figure 1A** [↗](#)) as indicated by the minimal bleeding near the bone flap (**Figure 1B** [↗](#)). During drilling, pressurized air and a cold stream of saline were used to remove the bone debris and dissipate heat. Because the IC sits deeper in the brain compared to the surrounding structures, the direct insertion of the 45° micropism was found to push the IC ventrally. Therefore, a fine diamond knife was used to separate the lateral surface of the IC from the surrounding tissues without damaging the transverse and sigmoid sinuses (**Figures 1C and D** [↗](#)). This step was followed by a gentle retraction of the knife medially to generate a pocket to secure the micropism (**Figure 1E** [↗](#)). The successful insertion of the micropism was assessed by examining the reflection of the diamond knife from the micropism's top glass surface (**Figures 1F and G** [↗](#)). Holding the micropism down and retracting the knife upward simultaneously resulted in securing the micropism in its pocket (**Figure 1H and I** [↗](#)). Locking the micropism down in its pocket for 10 minutes with the diamond knife (**Figure 1J** [↗](#)) is critical to ensure that the

microprism is stabilized and to prevent bleeding. Filling the gap between the skull surface and the microprism with saline was found to generate motion artifact due to the heart pulsation, so 1% agarose was used to fill the gap to reduce the motion artifact and add more stability to the microprism (**Figure 1K**). To compare response properties between the LC and DC, direct imaging (i.e., without a microprism) of the dorsal surface of the IC, where the DC is located, was also conducted by scanning the DC surface from the most medial to lateral regions following the convex curvature of the surface (**Figures 1L and M**).

The 2P characterization of GABAergic and non-GABAergic neuronal distribution on the LC and DC surfaces with histological validation

Following the surgery, the anesthetized animals were taken immediately for imaging. Examination of the distribution of GABAergic modules in the LC was critical to validate the target area for functional calcium imaging. Therefore, the 920 nm laser beam was initially used to scan for GFP, which is expressed in the GABAergic cells of GAD67-GFP knock-in mice. The insertion of the microprism was initially viewed by low magnification fluorescence imaging of the GFP signals from the whole surface of the IC along with the inserted microprism (**Figure 2A**).

This step validated the location of the microprism relative to the IC under the microscope and ensured there was no bleeding that could block the field of view. The 2P images of the GFP signals from the LC imaged via the microprism across different animals revealed the LC characteristic GABAergic modules (**Figures 2C and E**, the irregular white lines), which were consistent with previous 3D reconstructions of modules from the coronal sections (Chernock et al., 2004; A. M. Lesicko et al., 2016). To further validate the microprism location and the ability to image the LC from the lateral surface, a laser lesion (see Methods) was created via the microprism (**Figure 2G**), which was validated histologically (**Figure 2H**). For comparison, the whole dorsal surface of the IC was scanned from medial to lateral borders (**Figure 2B**, the dotted white lines). Unlike the LC, most of the dorsal surface of the IC showed a non-clustered distribution of the GABAergic cells (**Figures 2D and F**, blue boxes). However, the lateral horizon of the IC dorsal surface showed clusters of GABAergic cells and terminals that were confirmed to be modules, which comprise the prominent feature of the LC (**Figures 2D and F**, red boxes). To further validate the presence of GABAergic modules in the dorsolateral surface of the IC, a laser lesion was created in a GABAergic module located in that area (**Figures 2I and J**, dotted red circle), which was confirmed histologically (**Figure 2K**), demonstrating that it is possible to image a portion of the LC directly from its dorsal surface.

Using the 1040 nm laser, the GAD67-GFPx jRGECO1a hybrid mouse showed a non-clustered cellular expression of jRGECO1a on the surface of the DC (**Figure 3A**) or on the surface of the LC imaged via microprism (**Figure 3B**), as expected. The IC of anesthetized animals in each experimental setup was then imaged to measure calcium signals over the course of pseudorandom presentations of either 500 ms pure tones of different combinations of frequencies (5-40 kHz, half octave intervals) and sound levels (40-80 dB SPL, 10 dB intervals) or broadband noise without or with amplitude modulation (AM-noise) of different combinations of AM frequencies (2-256 Hz, one-octave intervals) and sound levels (40-80 dB SPL, 10 dB intervals). As shown by two representative cells from the DC imaged from the dorsal surface or LC imaged via microprism, these acoustic stimulation paradigms successfully elicited transient cellular calcium signals that varied depending on the stimulus type (**Figures 3C and D**).

Acoustic responsiveness differs in the LC and DC

As indicated by the GFP signals and consistent with **Figure 2**, imaging the LC of the hybrid mice via microprism showed clusters of GABAergic modules, which take different irregular shapes on the surface of the LC (**Figure 4A** - irregular white lines). Based on the best pure tone frequency

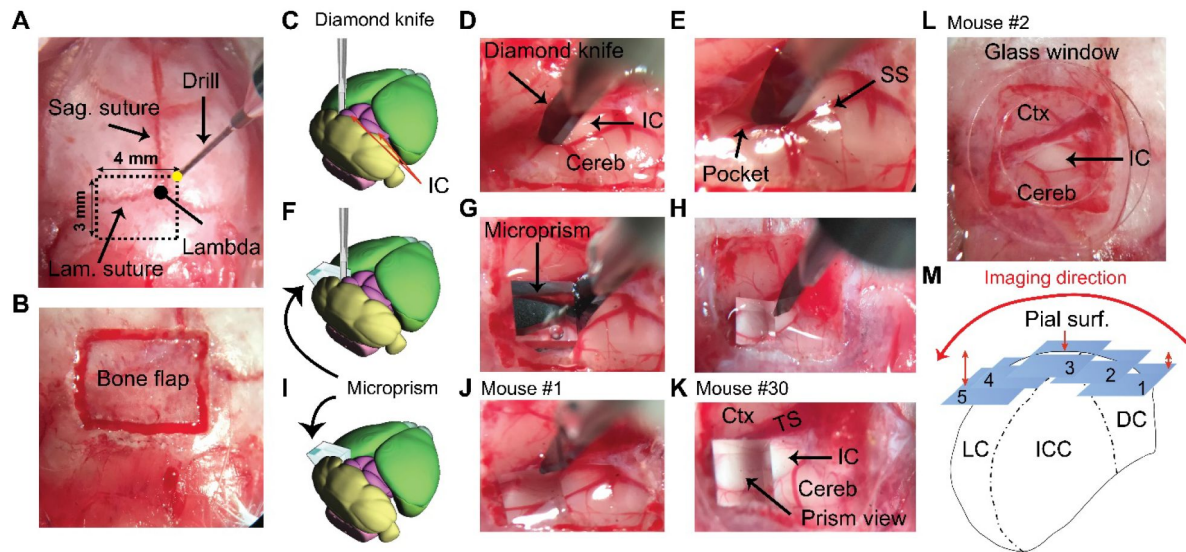


Figure 1

A brief illustration of the surgical procedures.

(A) A surface view of the mouse's skull showing the landmarks of the craniotomy over the IC. (B) An image of the bone flap after finishing the drilling. (C-E) A cartoon and two images of the brain surface showing the insertion and the pocket made by the diamond knife at the lateral surface of the left IC. (F-H) A cartoon and two images of the brain surface showing the insertion of the microprism into its pocket to directly face the LC after displacing the blade of the diamond knife. (I-K) A cartoon and two images of the brain surface showing the final setup of the microprism for two different animals. (L) An image of the brain surface showing the placement of a glass window for the surface imaging of the DC. (M) A cartoon of the coronal section of the IC demonstrating the different fields of views taken along the mediolateral axis parallel to the curvature of the IC surface at different depths (1-5) from the pial surface. Cereb: Cerebellum, Ctx: Cerebral cortex, DC: Dorsal cortex of the IC, IC: Inferior colliculus, ICC: Central nucleus of the IC, Lam. Lambdoid, LC: Lateral cortex of the IC, Sag. Sagittal, SS: Sigmoid sinus, Surf: Surface, TS: Transverse sinus.

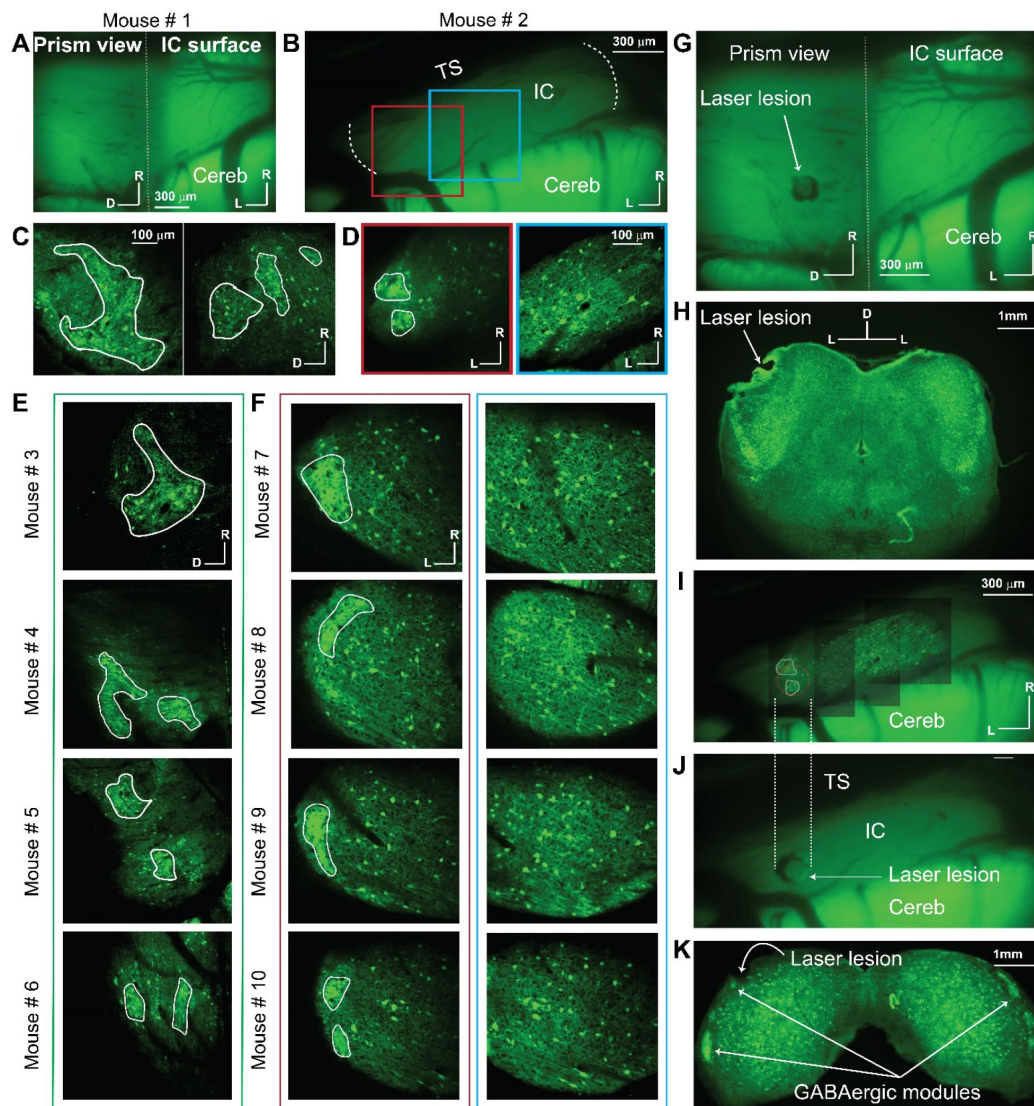


Figure 2

The distribution of GABAergic cells on the LC imaged via microprism, and DC imaged from the dorsal surface.

All images were taken for the GFP signals exclusively expressed in GABAergic cells. (A-B) Low magnification of fluorescence images for the GFP signals obtained from the IC surface with or without the microprism, respectively. The dotted white line in B represented the medial and lateral horizons of the IC. (C-E) The 2P images of the GABAergic cells on the LC imaged via microprism showing the GABAergic modules across multiple animals. (D and F) The 2P images of the GABAergic cells on either the dorsolateral surface of the DC imaged directly from its dorsal surface showing the GABAergic modules across multiple animals (red boxes) or its dorsomedial surface showing a homogenous distribution of GABAergic cells of different sizes with no signs of the GABAergic modules across multiple animals (blue boxes). (G) A low magnification fluorescence image of the IC surface with the microprism showing a laser lesion spot made through the microprism to validate the position of the microprism relative to the IC. (H) A histological coronal section at the level of the IC showing that the laser lesion made by the microprism was located at the sagittal surface of the LC imaged via microprism. (I-J) Low-magnification fluorescence images of the IC surface from the dorsal surface show a laser lesion spot made on the dorsolateral surface of the LC on the GABAergic module. The dotted white lines represent the alignment of the laser lesion (dotted red circle) to the location of the GABAergic module at the dorsolateral surface. (K) A histological coronal section at the level of the IC showing that the laser lesion made on the IC surface targeted the GABAergic module at the dorsolateral portion of the LC. All solid irregular white lines were made at the border of the GABAergic modules; D: Dorsal, IC: Inferior colliculus, L: Lateral, R: Rostral. Cereb: Cerebellum, TS: Transverse sinus.

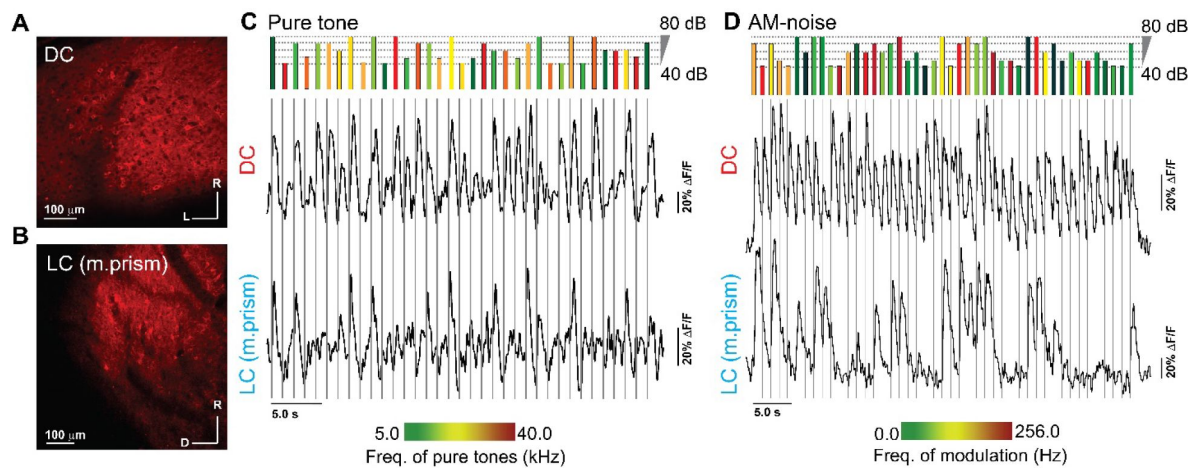


Figure 3

2P expression of jRGECO1a along with evoked signals in the DC imaged from the dorsal surface and LC imaged via microprism.

(A-B) The 2P images of the jRGECO1a expression on the DC imaged from the dorsal surface and the LC imaged via a microprism, respectively. (C-D) The time traces of the evoked calcium signals by either different frequencies and amplitudes of pure tones or different overall amplitudes and rates of AM of broadband noise, respectively, which were obtained from two different cells on the DC (top traces) or the LC imaged via microprism (bottom traces); The length of the colored bars (top) indicates the intensity of the stimulus, while the color indicates the tone or AM frequency depending on the color scale (bottom). AM-noise: Amplitude modulated broadband noise, D: Dorsal, DC: Dorsal cortex, L: Lateral, LC: Lateral cortex, m.prism: Microprism, R: Rostral.

of each responsive cell (Barnstedt et al., 2015 [\[4\]](#)), an inconsistent and fragmentary tonotopic organization was seen over the LC imaged via microprism (**Figure 4B** [\[4\]](#)). For comparison, the surface of the DC was imaged directly from the dorsal surface using the routine imaging setup. To view the majority of the DC surface, imaging was done by scanning the DC from its medial to the lateral horizon to generate a composite map (see Methods). As indicated by the GFP signals, it was noted that the most lateral region of the surface of the IC showed clusters of the GABAergic cells and terminals (i.e., modules, **Figure 4C** [\[4\]](#) – irregular white lines) and are therefore considered part of the LC. Based on the best pure tone frequency of each responsive cell (Barnstedt et al., 2015 [\[4\]](#)), the DC showed a tonotopic organization of cells along the rostromedial to caudolateral axis tuned to lower or higher frequencies, respectively (**Figure 4D** [\[4\]](#)). This tonotopic organization is similar to those previously described (Barnstedt et al., 2015 [\[4\]](#); A. B. Wong & Borst, 2019 [\[5\]](#)). It was also noticed that a large area of the DC was tuned to the lowest tested frequency (5kHz), suggesting a low frequency bias on the surface of the DC. To quantify tuning differences between DC and LC, the cumulative distribution function based on the cells' best frequency showed a low-frequency bias in the DC, similar to previous work (Ito et al., 2014 [\[6\]](#); Romand & Ehret, 1990 [\[7\]](#); Shen, Xu, & Shen, 2003 [\[8\]](#); Stiebler & Ehret, 1985 [\[9\]](#)), while most of the LC cells imaged via microprism were tuned to higher frequencies (**Figure 4E** [\[4\]](#), Mann-Whitney Test: $z = -13.4$, $p < 0.001$, a median of best-tuned frequency = 5 and 7.5 Hz for DC and LC via microprism, respectively).

Detection of AM is critical to identify natural sounds as well as to aggregate sound sources during auditory scene analysis, which is important for communication (Joris, Schreiner, & Rees, 2004 [\[10\]](#); Rees & Langner, 2005 [\[11\]](#); Singh & Theunissen, 2003 [\[12\]](#)). The IC also is the first site in the ascending auditory pathway where a substantial number of neurons are rate-tuned to AM (Frisina, 2001 [\[13\]](#); Joris et al., 2004 [\[10\]](#); Langner, 1992 [\[14\]](#); Nelson & Carney, 2007 [\[15\]](#); Rabang et al., 2012 [\[16\]](#)). Even though calcium indicators do not provide a spike-by-spike readout of neural activity, their integrated signal does correlate with the number of spikes produced by a neuron over short time windows (Vogelstein et al., 2010 [\[17\]](#); Yaksi & Friedrich, 2006 [\[18\]](#)). Thus, this integrated signal can be used to measure rate responses to AM. Using this integrated signal, we tested the ability of the cells of LC imaged via microprism in its different motifs to respond to broadband noise at different AM rates. We observed a significant fraction of cells that did not respond to pure tones responded to AM-noise (**Figure 5A** [\[4\]](#), paired-sample t-test: $t_{(6)} = -11.13$, $p = 3.1 \times 10^{-5}$, the percentage of responsive cells mean $\pm$ SEM = $34.7 \pm 9.2\%$ and $56.2 \pm 8.9\%$ for pure tones and AM-noise, respectively). We also observed that modules and matrix differentially process the temporal information. For instance, the matrix had a significantly greater fraction of cells that respond only to AM-noise compared to the modules (**Figure 5B** [\[4\]](#) – striped blue bar, Mann-Whitney Test, $p = 0.02$), which suggests that neurons in the matrix are more sensitive to complex sounds compared to those in the modules. In contrast, both cellular motifs had similar fractions of cells that are responsive to pure tones (**Figure 5B** [\[4\]](#) – solid blue bar, Mann-Whitney Test, $p = 0.58$). These findings suggest that neurons in the matrix are more responsive to sounds of greater spectral and temporal complexity than those in the modules. In the DC, like the LC imaged via microprism, a significant fraction of cells that did not respond to pure tones did respond to AM-noise (**Figure 5C** [\[4\]](#), paired samples t-test: $t_{(3)} = -5.8$, $p = 0.01$, the percentage of responsive cells mean $\pm$ SEM = $78.4 \pm 3.6\%$ and $97.3 \pm 0.4\%$ for pure tones and AM-noise).

Regardless of the sound type, animals showed a lower fraction of responsive neurons to the acoustic stimulation in the LC imaged via microprism compared to DC (pure tone: two-sample t-test, $t_{(9)} = 2.9$, $p = 0.008$, the percentage of responsive cells mean $\pm$ SEM = $78.4 \pm 3.6\%$ and $34.7 \pm 9.2\%$ for DC and LC, respectively, AM-noise: two-sample t-test, $t_{(9)} = 3.5$, $p = 0.008$, the percentage of responsive cells mean $\pm$ SEM = $97.3 \pm 0.4\%$ and $56.2 \pm 8.9\%$ for DC and LC, respectively). To evaluate if any optical artifacts due to imaging through a prism could be responsible for this finding, the number of responsive cells imaged from the lateral-most dorsal surface of the IC, which we consider part of the LC because of the presence of GABAergic modules (LC imaged from the dorsal surface), was quantified and compared with that obtained from the LC

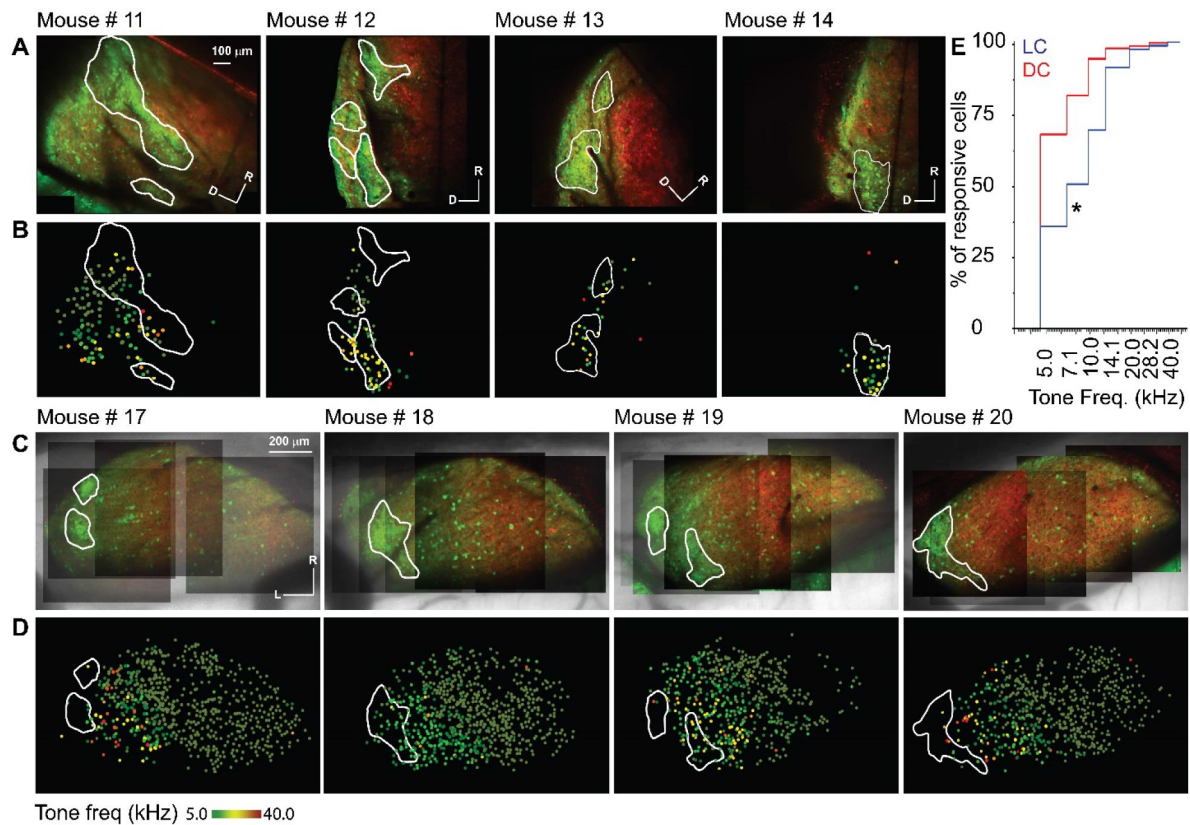


Figure 4

The auditory cellular organization of the LC imaged via microprism vs. DC imaged from the dorsal surface.

(A and C) The 2P images of GFP (green) and jRGECO1a expression (red) on either the LC imaged via the microprism or the DC imaged from the dorsal surface, respectively. (B and D) The pseudocolor images show the responsive cells to the pure tone of different combinations of frequencies and levels on the surface of the LC imaged via the microprism or the DC imaged from its dorsal surface, respectively. (E) The cumulative distribution function of the best tone frequencies of all cells collected either from the LC imaged via microprism (blue line) or the DC imaged directly from its dorsal surface (red line). D: Dorsal, L: Lateral, R: Rostral. (* $p < 0.05$, LC vs. DC).

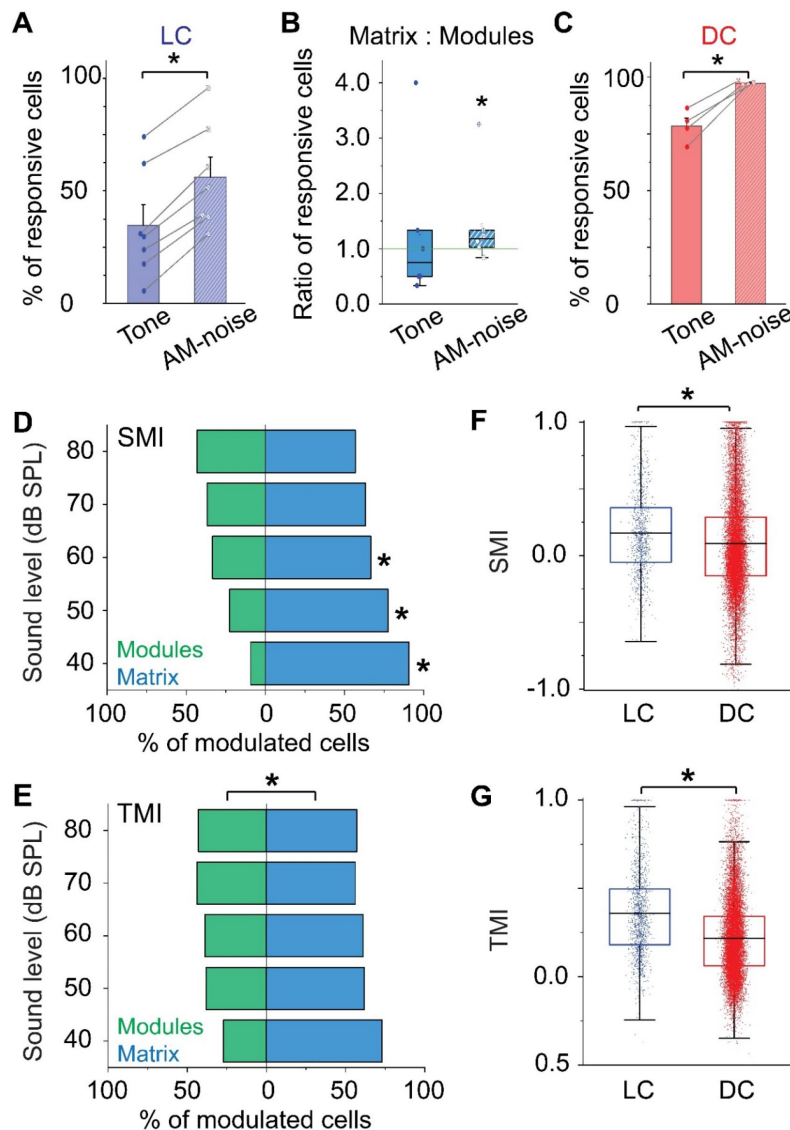


Figure 5

The cellular response to pure tones and AM-noise of the LC imaged via microprism vs. DC imaged from the dorsal surface.

(A and C) Bar graphs showing the difference in the cellular response to pure tone (tone) vs. AM-noise on the LC imaged via microprism or the DC imaged directly from the dorsal surface, respectively (* $p < 0.05$, pure tone vs. AM-noise). (B) A bar graph showing the difference in the cellular response to the pure tone (tone) and AM-noise of the matrix after normalization against modules imaged via microprism (* $p < 0.05$, matrix vs. modules). (D and E) Bar graphs showing the percentage of spectrally (two-way ANOVA: $f_{(1, 4, 9)} = 50.3, 0.0, \text{ and } 4.0 - p = 7.1 \times 10^{-7}, 1.0, \text{ and } 0.01$ for the region, sound level, and interaction, respectively, Fisher post hoc, $p = 5.1 \times 10^{-6}, 4.9 \times 10^{-4}, 0.02, 0.058, \text{ and } 0.14$ for modules vs. matrix at 40, 50, 60, 70, and 80 dB, respectively) or temporally (two-way ANOVA: $f_{(1, 4, 9)} = 19.3, 0.0, \text{ and } 1.2 - p = 7.1 \times 10^{-7}, 1.0, \text{ and } 0.32$ for the region, sound level, and interaction, respectively, Fisher post hoc, $p = 2.7 \times 10^{-7}$ for modules vs. matrix) modulated cells, respectively, across different sound levels in modules (green bars) vs. matrix (blue bars) imaged via microprism. (* $p < 0.05$, matrix vs. modules). (F and G) box plots showing the mean (black lines) and the distribution (colored dots) of the SMI (two-way ANOVA: $f_{(1, 4, 14583)} = 48.4, 210.4, \text{ and } 21.3 - p < 0.001$ for the region, sound level, and interaction, respectively, Fisher post hoc test, $p = 6.2 \times 10^{-4}, 4.2 \times 10^{-5}, < 0.001, \text{ and } < 0.001$ for LC vs. DC at 40, 60, 70, and 80 dB respectively) and TMI (two way ANOVA: $f_{(1, 4, 16139)} = 431.4, 59.1, \text{ and } 8.3 - p < 0.001$ for the region, sound level, and interaction, respectively, Fisher post hoc test, $p = 0.002, < 0.001, < 0.001, < 0.001, \text{ and } < 0.001$ for LC vs. DC at 40, 50, 60, 70, and 80 dB, respectively), respectively, across different sound levels in LC imaged via microprism (blue) vs. DC imaged directly from the dorsal surface (red) (* $p < 0.05$, LC vs. DC). LC: Lateral cortex, DC: Dorsal cortex.

imaged via microprism. We observed that there were no significant differences between the number of tone or noise-responsive neurons between the LC imaged via microprism or directly from the dorsal surface (pure tone: two-sample t-test, $t_{(9)} = 1.1$, $p = 0.28$, AM-noise: two-sample t-test, $t_{(9)} = 1.6$, $p = 0.14$).

LC and DC respond differently to spectral and temporal modulation of acoustic inputs

Given that more cells from LC imaged via microprism and DC had responses to AM-noise compared to pure tones, we further quantified the selectivity to spectral or temporal modulation by computing the spectral modulation index (SMI) or the temporal modulation index (TMI), respectively, for all the responsive cells (see Methods) across the tested sound levels.

For the LC imaged via microprism, the LC matrix has a greater proportion of spectrally modulated cells than modules most prominently at sound levels below 70 dB (**Figure 5D**, two-way ANOVA: $p = 7.1 \times 10^{-7}$, 1.0, and 0.01 for the region, sound level, and interaction, respectively, the percentage of modulated cells mean $\pm$ SEM for modules vs. matrix) suggesting that the cells in the matrix have a lower threshold to detect complex spectral features of sound. Similarly, the matrix had more cells responsive to temporal modulation as a main effect (**Figure 5E**, two-way ANOVA, $p = 7.1 \times 10^{-7}$, 1.0, and 0.32 for the region, sound level, and interaction, respectively), which was consistent with the above results (**Figure 5B**).

The DC and LC imaged via microprism were found to have different profiles of modulation of responses to spectral and temporal complexity. Although there is a general trend for SMI and TMI to decrease with increasing sound-pressure level, corresponding to an increase in pure-tone responsiveness, overall there is a statistically-significant higher level of average SMI and TMI in the LC compared to the DC (**Figures 5F and G**, SMI: two-way ANOVA, $p < 0.001$ for all region, sound level, and interaction, TMI: two-way ANOVA, $p < 0.001$ for all regions, sound level, and interactions). These results suggested that neurons in the LC are more sensitive to spectral and temporal modulation than those in the DC.

LC modules and matrix differentially respond to acoustic and somatosensory stimuli

Somatosensory inputs from many cortical and subcortical structures have been identified in the non-lemniscal division of the IC, particularly in the LC (A. M. Lesicko et al., 2016; Lohse, Zimmer-Harwood, Dahmen, & King, 2022; Zhou & Shore, 2006). Previous work has shown that while somatosensory projections mostly target the modules, the auditory projections tend to target the matrix (A. M. Lesicko et al., 2016), which indicates that the LC is a multisensory integration site. Therefore, we examined the somatosensory and auditory cellular responses on the surface of the LC by coupling the 2P imaging with the microprism with a whisker or acoustic stimulation (see Methods). Acoustic stimulation was done with 500 ms broadband noise at 80 dB SPL. After the acoustic or somatosensory stimulation, evoked cellular responses were observed from the LC neurons imaged via microprism either at the beginning of the stimulation (**Figures 6A and B** – onset responses) or at the termination of the stimulation (**Figures 6C and D** – offset responses).

Auditory and somatosensory responses were evoked in the neurons of both matrix and modules imaged via microprism (**Figures 7A, B, and D**) and the cells were categorized within each motif into cells that respond only to acoustic stimulation, only to somatosensory stimulation, or to both stimuli (**Figure 7F**). The quantification of the percentage of cells in both modules and matrix revealed that the percentage of responsive cells to sound only was higher than that of cells responsive to somatosensory stimulation or those responsive to both stimuli with no preference (Modules: **Figure 7H**, one-way ANOVA, $p = 4.0 \times 10^{-5}$, the percentage of responsive cells mean $\pm$ SEM = $41.9 \pm 5.4\%$ (auditory-only response), $4.2 \pm 0.8\%$ (somatosensory-only response), and $10.2 \pm$

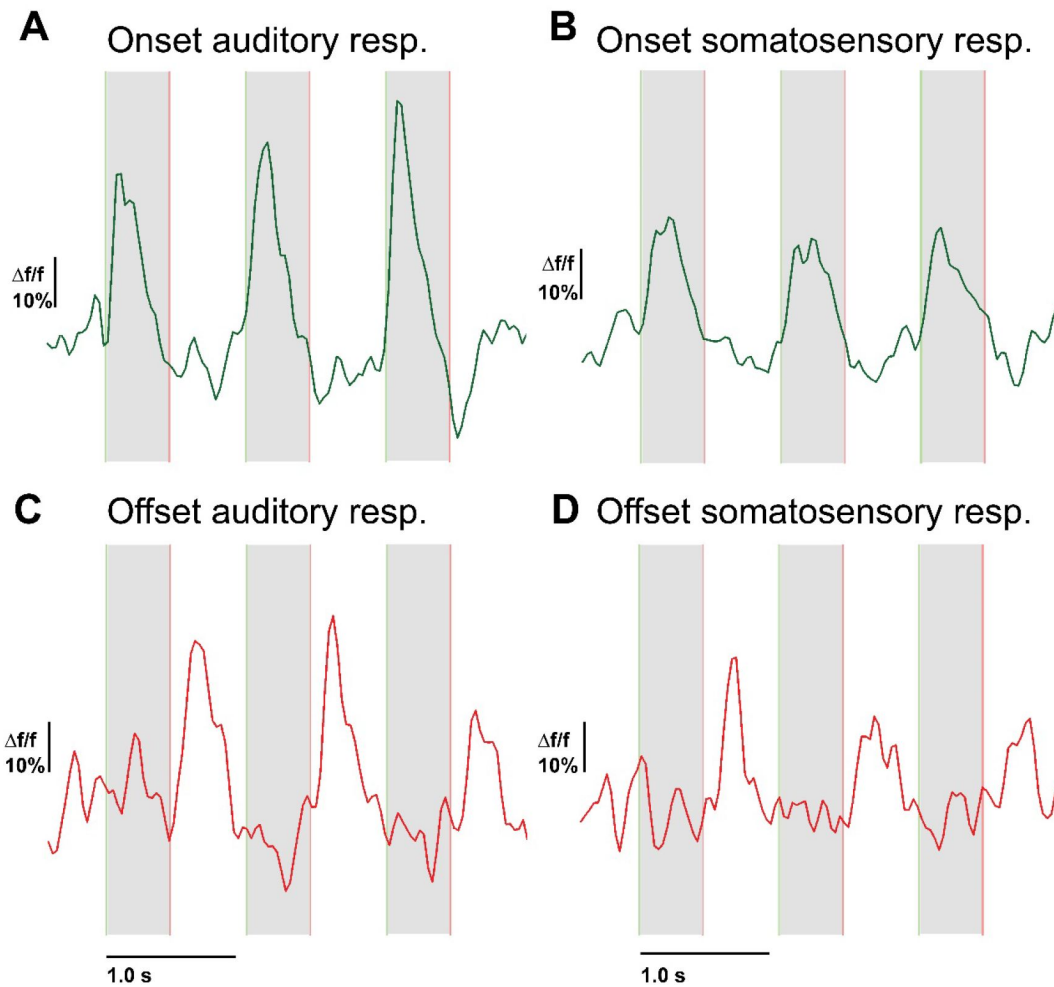


Figure 6

The waveforms of the somatosensory vs. acoustic responses of the LC imaged via microprism.

(A-B) The time traces of the evoked calcium signals by the beginning of auditory or somatosensory stimuli, respectively (onset responses). (C-D) The time traces of the evoked calcium signals by the termination of auditory or somatosensory stimuli, respectively (offset responses).

1.4% (auditory + somatosensory responses)) (Matrix: **Figure 7I**, One-way ANOVA, $p = 2.4 \times 10^{-4}$, the percentage of responsive cells mean $\pm$ SEM = $43.7 \pm 8.5\%$ (auditory response), $0.88 \pm 0.36\%$ (somatosensory response) and $2.9 \pm 0.34\%$ (auditory + somatosensory responses)). However within the pool of responsive cells and normalized against modules, we observed that more sound-responsive cells were found in the matrix, while more somatosensory responsive cells were found in the modules (**Figures 7B**, D, F, and J, Mann-Whitney Test, $p = 0.028$), which was consistent with the reported difference in inputs to modules compared to matrix (A. M. Lesicko et al., 2016). Most of the acoustically-responsive cells in the matrix showed onset responses, and the cells showing acoustically-driven offset responses were clustered in the modules (**Figures 7C and K**, Mann-Whitney Test, $p = 0.028$).

Additionally, most of these auditory-offset cells were also responsive to somatosensory stimulation as shown by the merged responses (**Figure 7F**), which suggests a multisensory integrative function by those cells as discussed later. Given the specific spatial location of the somatosensory-responsive cells in the modules which contain clusters of GABAergic cells, we asked if somatosensory stimulation could suppress the auditory response when both stimuli are presented simultaneously (bimodal stimulation). To test this hypothesis, we calculated and visualized the response index (RI), which was calculated as the ratio of the cellular response to bimodal stimulation to the cellular response to acoustic response alone for auditory-responsive cells. We observed that cells in the modules had a higher odds of having an enhancement response to bimodal stimulation than matrix, while cells in the matrix had a higher odds of having a suppressive response to bimodal stimulation (**Figures 7G and L**, enhanced response: Mann-Whitney Test, $p = 0.028$, matrix median odds = 0.68, suppressed response: Mann-Whitney Test, $p = 0.028$, matrix median odds = 1.33). These data suggest that the module/matrix system permits preservation of distinct multimodal response properties in the face of massive integration of inputs in the LC.

Consistent with the above data, the LC imaged from the dorsal surface (**Figures 8A-D**, dorsolateral) shared common response profiles with the LC. For instance, the LC imaged via microprism or directly from the dorsal surface had a higher percentage of somatosensory responsive cells than the DC (**Figure 8A-D**, dorsal and dorsomedial) as quantitatively shown (**Figure 8H**, one-way ANOVA, $p = 0.02$, the percentage of somatosensory responsive cells mean $\pm$ SEM = $1.3 \pm 0.83\%$ (DC), $8.2 \pm 2.4\%$ (LC imaged from the dorsal surface), and $8.1 \pm 0.64\%$ (LC imaged via microprism), which suggest that the LC is the main region of the IC shell that processes somatosensory information, consistent with the anatomical reports (Aitkin, Dickhaus, Schult, & Zimmermann, 1978; Jain & Shore, 2006; A. M. Lesicko et al., 2016; Liu et al., 2023). The matrix of the LC imaged from the dorsal surface had more responsive cells to sound (**Figures 8E and I**, Mann-Whitney Test, $p = 0.028$, the matrix:module median ratio = 1.5), and less responsive cells to somatosensory stimulation compared to its modules (**Figures 8E and I**, Mann-Whitney Test, $p = 0.028$, matrix:module median ratio = 0.28). Additionally, more acoustically onset responsive cells were found in the matrix (**Figures 8F and J**, Mann-Whitney Test, $p = 0.028$, matrix:module median ratio = 10.2), while more acoustically offset responsive cells were found in the modules (**Figures 8F and J**, Mann-Whitney Test, $p = 0.028$, matrix:module median ratio = 0.02). More acoustically responsive cells in the matrix displayed suppressed responses following the bimodal stimulation (**Figures 8G and K**, Mann-Whitney Test, & $p = 0.028$, matrix:module median ratio = 1.5). In contrast, more acoustically responsive cells in the modules displayed enhanced responses following the bimodal stimulation (**Figures 8G and K**, Mann-Whitney Test, & $p = 0.028$, matrix:module median ratio = 0.42). These data confirmed the findings obtained from the LC imaged via microprism and emphasized the functional difference between the LC and DC.

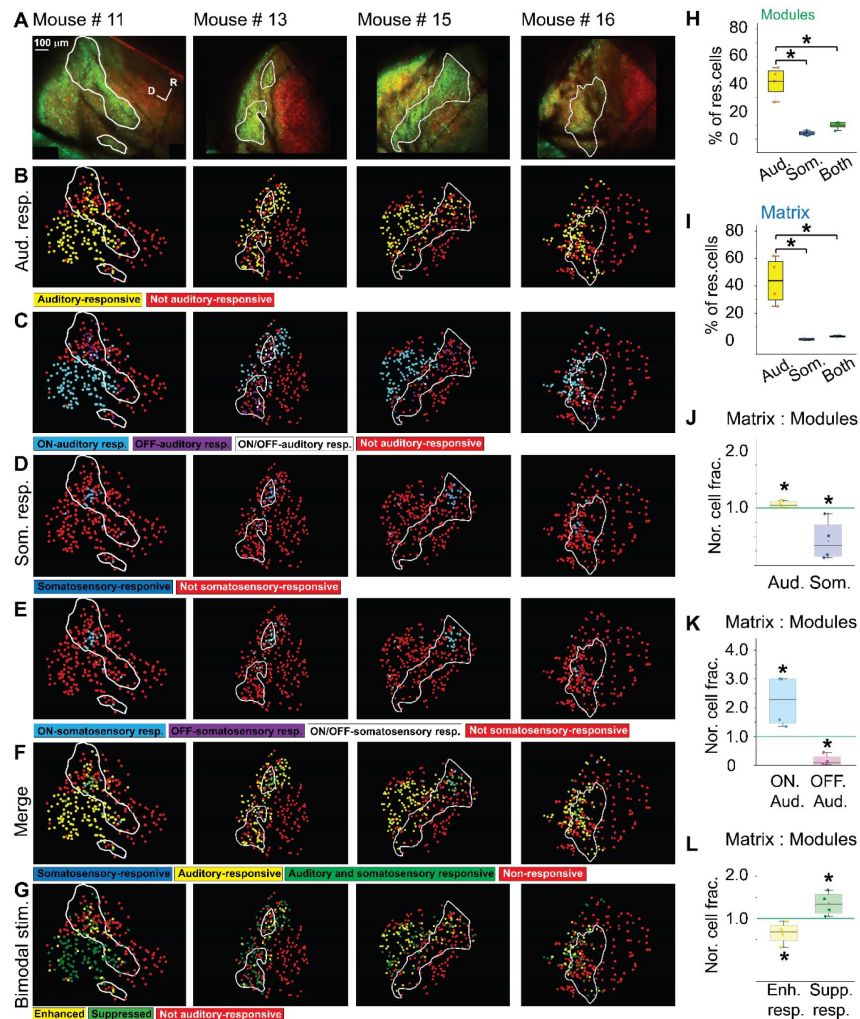


Figure 7

Types of somatosensory and auditory responses from the LC neurons imaged via microprism.

(A) The 2P images of GFP (green) and jRGECO1a expression (red) in the LC imaged via microprism showing the modules within white irregular lines. (B, D, and F) The pseudocolor images show the auditory or somatosensory responsive cells of the LC imaged via microprism as well as the merged response, respectively (C and E) The pseudocolor images show onset (ON), offset (OFF), or onset/offset (ON/OFF) responsive cells of the LC imaged via microprism to auditory and somatosensory stimuli, respectively. (G) The pseudocolor images show enhanced and suppressive response of the auditory responsive cells of the LC imaged via microprism following the bimodal stimulation based on their response index (RI). (H and I) Bar graphs showing the percentage of responsive cells to auditory (Aud.), somatosensory (Som.), or both stimuli in modules and matrix imaged via microprism, respectively – Modules: one-way ANOVA: $f_{(2, 11)} = 38.1$ - $p = 4.0 \times 10^{-5}$, Fisher posthoc test, $p = 1.9 \times 10^{-5}$ or 7.6×10^{-5} for the percentage of responsive cells to sound only vs. the percentage of responsive cells to somatosensory stimulation only or to both, respectively and $p = 0.23$ for the percentage of responsive cells to somatosensory stimulation only vs. both, Matrix: one-way ANOVA: $f_{(2, 11)} = 23.9$ - $p = 2.4 \times 10^{-4}$, Fisher posthoc test, $p = 1.7 \times 10^{-4}$ or 2.4×10^{-4} for the percentage of responsive cells to sound only vs. the percentage of responsive cells to somatosensory stimulation only or to both, respectively - $p = 0.77$ for the percentage of responsive cells to somatosensory stimulation only vs. both) (* $p < 0.05$, vs. Aud.). (J) A box plot showing the normalized cell fraction of auditory (Aud.) (matrix median = 1.05) or somatosensory (Som.) (matrix median = 0.14) responsive cells in the matrix normalized to those in modules imaged via microprism (* $p < 0.05$, matrix vs. modules). (K) A box plot showing the cell fraction of onset (ON. Aud.) (matrix median = 2.2) or offset (OFF. Aud.) auditory (matrix median = 0.08) responsive cells in the matrix normalized to those in modules imaged via microprism (* $p < 0.05$, matrix vs. modules). (L) A box plot showing the cell fraction of the responsive cells of enhanced (Enh. resp.) or suppressive (Supp. resp.) response in the matrix normalized to those in modules imaged via microprism following bimodal stimulation (* $p < 0.05$, matrix vs. modules). D: Dorsal, R: Rostral.

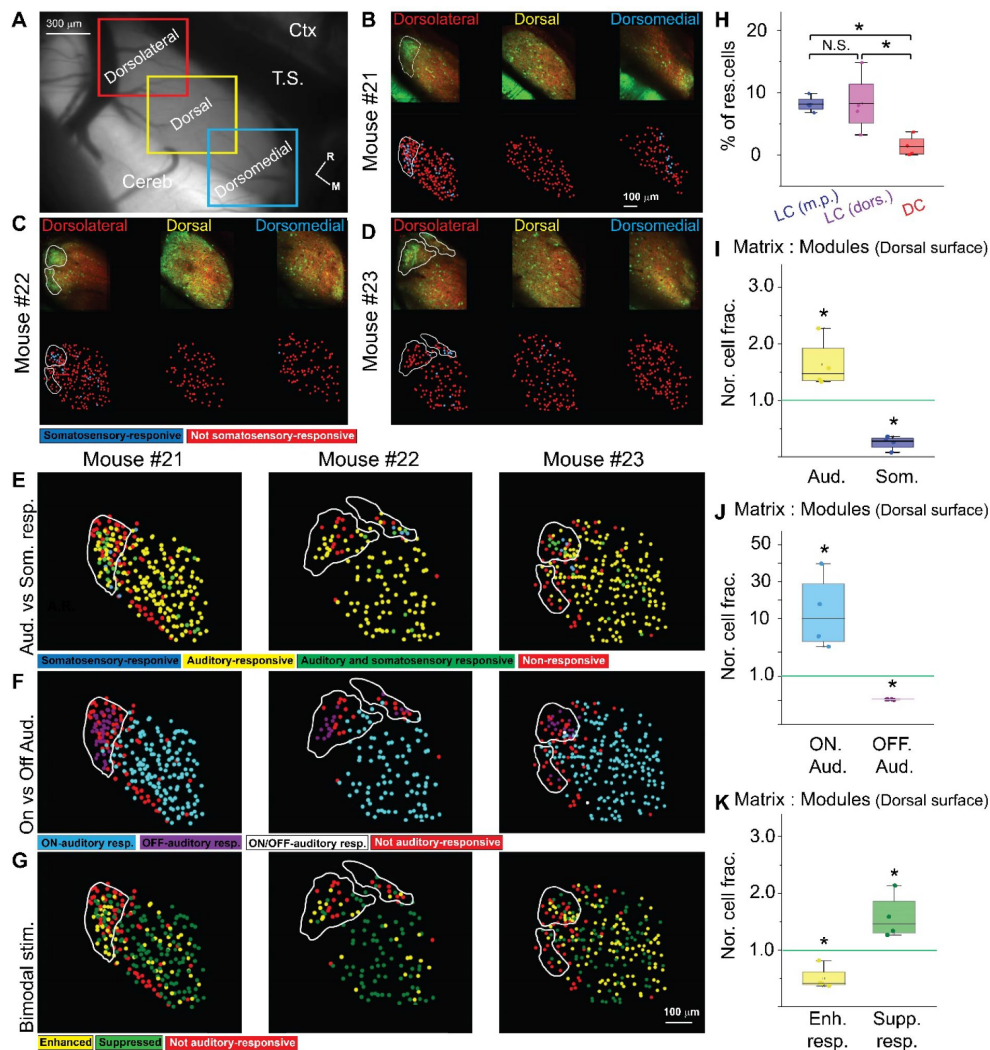


Figure 8

The somatosensory vs. auditory responses over the DC neurons.

(A) A low magnification of the image obtained from the IC surface. (B-D) The top row: The 2P images of GFP (green) and jRGECO1a expression (red) of three main regions (dorsolateral, dorsal, and dorsomedial) of the DC surface imaged from the dorsal surface showing the modules within irregular white lines in the dorsolateral region, the bottom row: The pseudocolor images show the somatosensory responsive cells after somatosensory stimulations. (E) The pseudocolor images show the responsive cells of the LC imaged from the dorsal surface to auditory and somatosensory stimulations. (F) The pseudocolor images show onset (ON), offset (OFF), or onset/offset (ON/OFF) responsive cells of the LC imaged from the dorsal surface to auditory stimulation. (G) The pseudocolor images show the enhanced and suppressive response of the auditory responsive cells of the LC imaged from the dorsal surface following the bimodal stimulation based on their response index (RI). (H) A box plot shows the percentage of somatosensory responsive cells in LC imaged either from the dorsal surface (LC (dors.)) or via microprism (LC (m. p.)) as well as the DC - one-way ANOVA: $f_{(2, 11)} = 6.7$ & $p = 0.02$, Fisher posthoc test, $p = 0.01$ or 0.01 for the percentage of somatosensory responsive cells on the DC vs. LC imaged by microprism or LC imaged from the dorsal surface, respectively - $p = 0.95$ for the percentage of somatosensory responsive cells on LC imaged via microprism vs. LC imaged from the dorsal surface ($*p < 0.05$, vs. DC). (I) A box plot showing the cell fraction of auditory (Aud.) or somatosensory (Som.) responsive cells in the matrix normalized to those in modules imaged from the dorsal surface ($*p < 0.05$, matrix vs. modules). (J) A box plot showing the cell fraction of onset (ON. Aud.) or offset (OFF. Aud.) auditory responsive cells in the matrix normalized to those in modules imaged from the dorsal surface ($*p < 0.05$, matrix vs. modules). (K) A box plot showing the cell fraction of the responsive cells of enhanced (Enh. Resp.) or suppressed (Supp. Resp.) responses in the matrix normalized to those in modules imaged from the dorsal surface following bimodal stimulation ($*p < 0.05$, matrix vs. modules). Cereb: Cerebellum, Ctx: Cerebral cortex, M: medial, R: Rostral, TS: Transverse sinus.

LC and DC differ in their acoustic and multimodal response properties in awake animals

To determine if the presence of ketamine/xylazine anesthesia influenced the response properties detailed above, additional experiments were performed in awake animals ($n=6$ total, 3 from LC using prism and 3 from dorsal surface). Similar to what was observed under anesthesia, the cells of the LC imaged via microprism did not show clear evidence for tonotopy (**Figure 9B**). In contrast, the data obtained from imaging the DC under awake preparations showed a similar tonotopic map (**Figure 9D**) as that observed under anesthesia, though a greater proportion of high-frequency cells were observed leading to the observation of a low-frequency border similar to the previous under awake preparations (A. B. Wong & Borst, 2019). Similar to the results obtained under anesthesia, the LC imaged via microprism had more cells tuned to higher frequencies compared to the DC (**Figure 9E**, Mann-Whitney test, $p=1.6\times 10^{-6}$, a median of best-tuned frequency = 10 and 5 kHz for LC and DC, respectively). Both LC imaged via microprism and DC imaged directly from its dorsal surface were more responsive to AM-noise than pure tones (**Figure 9F**, DC: χ^2 test = 15.1, $p = 0.001$, the percentage of responsive cells = 72.3 vs. 78.2% for pure tone vs. AM-noise, respectively, LC: χ^2 test = 43.6, $p = 1.8\times 10^{-9}$, the percentage of responsive cells = 58.7 vs. 76.1% for pure tone vs. AM-noise, respectively). Within each stimulus type, while the DC was found to have more responding cells to pure tone than LC imaged via microprism (**Figure 9G**, χ^2 test = 39.5, $p = 1.3\times 10^{-8}$, the percentage of responsive cells = 72.3 vs. 58.7% for DC vs. LC, respectively), both areas showed similar percentage of responsive cells to AM-noise (**Figure 9G**, χ^2 test = 1.2, $p = 0.75$, the percentage of responsive cells = 78.2 vs. 76.1 % for DC vs. LC, respectively, respectively). Additionally, the matrix also showed more responsive cells to sound compared to the modules regardless of the stimulus type (**Figure 9H**, pure tone: χ^2 test = 31.9, $p = 5.2\times 10^{-7}$, the percentage of responsive cells = 30.8 vs. 69.2% for modules vs. matrix, respectively, AM-noise: χ^2 test = 11.1, $p = 0.01$, the percentage of responsive cells = 37.6 vs. 62.4% for modules vs. matrix, respectively).

For somatosensory responses, similar to the anesthetized preparation, neurons in modules imaged via microprism had more cells responsive to somatosensory stimulation than the matrix, while the latter had more responsive cells to broadband noise (**Figures 10A**, C, E, and G, auditory response: χ^2 test = 31.9, $p = 5.2\times 10^{-7}$, the percentage of responding cells = 22 vs. 78% for modules vs. matrix, respectively, somatosensory response: χ^2 test = 11.1, $p = 0.01$, the percentage of responding cells = 63.6 vs. 36.4% for modules vs. matrix, respectively). Similar to anesthetized data, we found that most of the auditory responsive cells in the modules and matrix had suppressed auditory responses following bimodal stimulation (**Figures 10F and H**, χ^2 test = 31.9, $p = 5.2\times 10^{-7}$, matrix: The percentage of responding cells = 69.6 vs. 30.4% for suppressed vs. enhanced responses, respectively, modules: The percentage of responding cells = 95 vs. 5% for suppressed vs. enhanced responses, respectively), indicating an overall suppression in the cellular auditory response.

Discussion

Summary

As a new imaging approach, a microprism was used to image the LC surface, to provide the first-ever sagittal image of the LC in vivo and to examine its functional organization, which revealed the distinction between strategies used by LC and DC for processing acoustic and non-acoustic information. Imaging the LC either via microprism or from the dorsal surface revealed that it contains GABAergic modules that were embedded in mostly non-GABAergic matrix. Although modules and matrix were found to be activated by somatosensory and auditory stimuli, they were functionally distinct. The matrix had more auditory-responsive cells with a lower threshold for complex sound detection and had more cells responsive to complex spectral and temporal

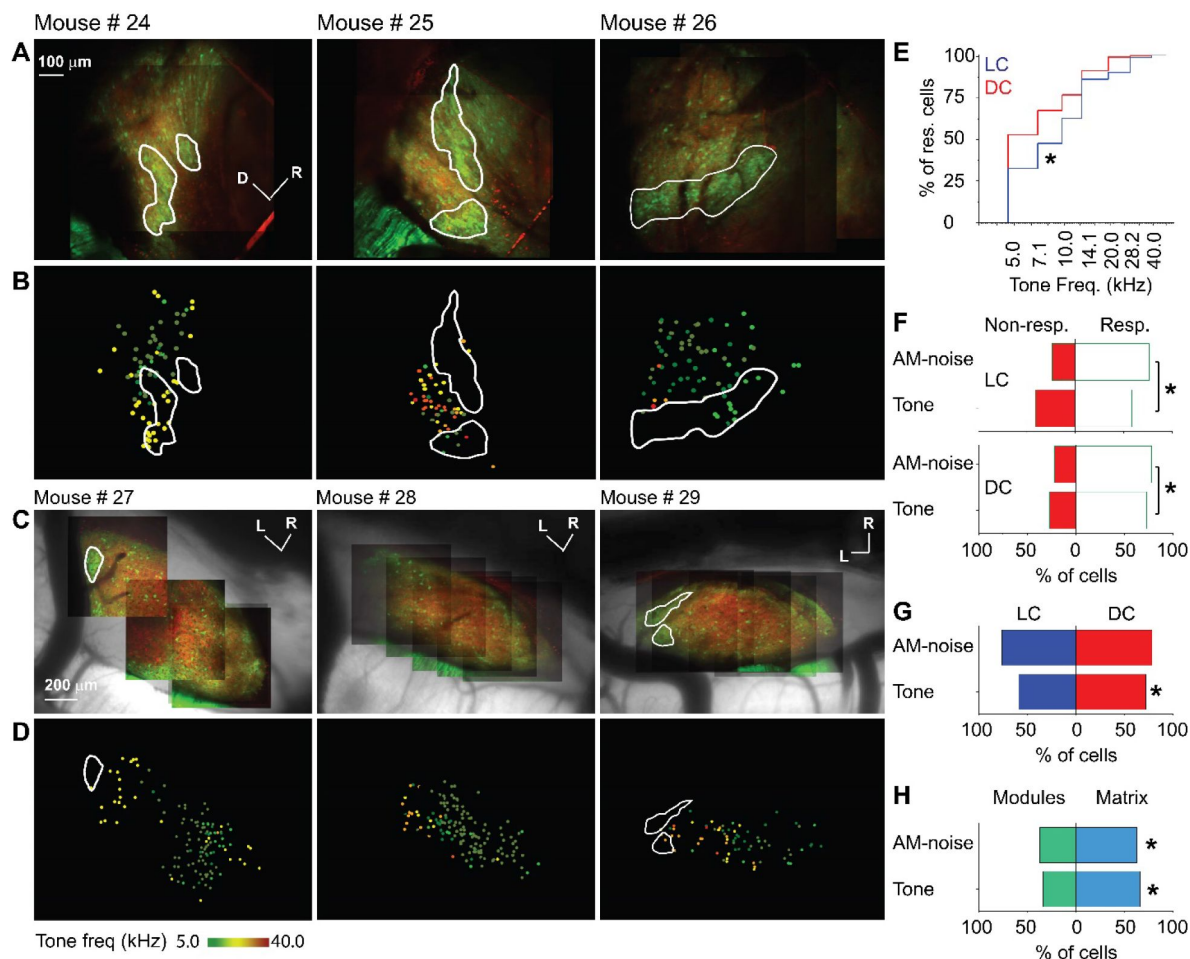


Figure 9

The acoustic response from LC imaged via microprism and DC imaged from the dorsal surface in awake animals.

(A and C) The 2P images of GFP (green) and jRGECO1a expression (red) on either the LC surface imaged via the microprism or the DC imaged directly from the dorsal surface, respectively showing the modules within irregular white lines. (B and D) The pseudocolor images show the responsive cells to the pure tone of different combinations of frequencies and levels of the LC imaged via the microprism or the DC imaged directly from the dorsal surface, respectively. (E) The cumulative distribution function of the best tone frequencies of all cells collected either from the LC imaged via microprism (blue) or the DC (red). (F) A bar graph showing the percentage of responsive vs. nonresponsive cells to pure tone or AM-noise within the LC imaged via microprism or the DC (* $p < 0.05$, pure tone vs. AM-noise). (G) A bar graph showing the percentage of responsive cells to pure tone or AM-noise for LC imaged via microprism vs. DC (* $p < 0.05$, LC vs. DC). (H) A bar graph showing the percentage of responsive cells to pure tone or AM-noise for modules vs. matrix imaged via microprism (* $p < 0.05$, matrix vs. modules). D: Dorsal, L: Lateral, Non-resp.: Non-responsive, Resp.: Responsive, R: Rostral.

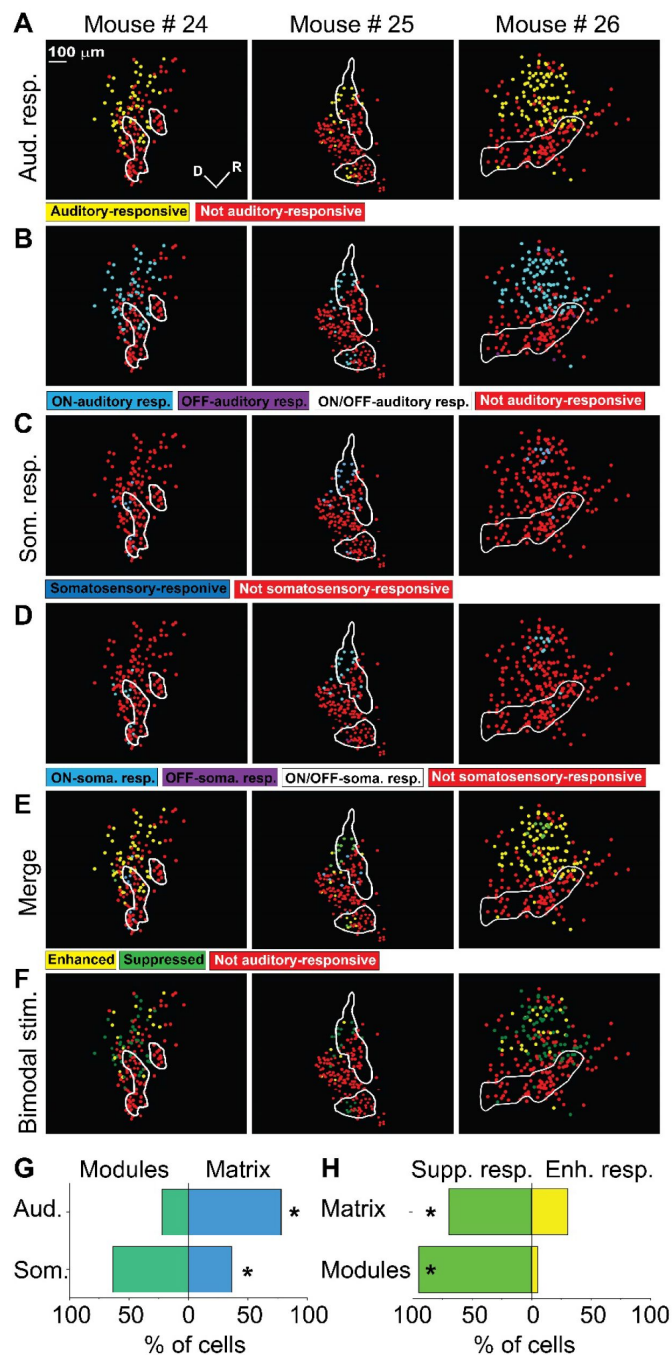


Figure 10

The somatosensory response from LC imaged via microprism in awake animals.

(A, C, and E) The pseudocolor images show the auditory or somatosensory responsive cells of the LC imaged via microprism to auditory and somatosensory stimulations as well as the merged response, respectively. (B and D) The pseudocolor images show onset (ON), offset (OFF), or onset/offset (ON/OFF) responsive cells of the LC imaged via microprism to auditory and somatosensory stimulations, respectively. (F) The pseudocolor images show the enhanced or suppressed responses of the auditory responsive cells following the bimodal stimulation based on their response index (RI). (G) A bar graph showing the cell fraction of auditory (Aud.) or somatosensory (Som.) responsive cells in the matrix vs. modules imaged via microprism (* $p < 0.05$, matrix vs. modules). (H) A bar graph showing the cell fraction of the responsive cells of suppressed (Supp. resp.) or enhanced (Enh. resp.) response in the matrix vs. modules imaged via microprism following bimodal stimulation (* $p < 0.05$, low vs. high response). D: Dorsal, R: Rostral.

auditory features than modules. In contrast, modules had more somatosensory responsive cells, consistent with anatomical differences in input to the modules compared to matrix (A. M. Lesicko et al., 2016). Somatosensory stimulation suppressed auditory responses in the matrix, suggesting that multisensory interactions exist between the matrix and modules.

The novelty of the imaging setup and technical considerations

Microprism-based 2P-microscopy has been successfully used to image the cortical layers in the somatosensory and visual cortices of mice (Andermann et al., 2013) and to image other inaccessible cortical surfaces like those folded in the brain sulci such as the medial prefrontal cortex (Low, Gu, & Tank, 2014) or entorhinal cortex (Beckmann et al., 2019; Heys, Rangarajan, & Dombeck, 2014; Low et al., 2014). In those studies, the glass window glued to the microprism was used, which requires significant skill and potential removal of brain regions to allow prism access (Heys et al., 2014). Another approach is to use a micro metal or diamond-made knife to create a pocket with minimal tissue damage (Andermann et al., 2013; Marks & Goard, 2021).

Accordingly, we combined and modified some of these procedures to improve the microprism implantation to image the sagittal surface of the LC, which is deeply embedded into the brain tissue in a highly vascular area, with minimum damage to the IC or the surrounding vasculature.

Previous work using microprisms has been done chronically for up to 60 days (Heys et al., 2014; Low et al., 2014; Wenzel, Hamm, Peterka, & Yuste, 2017), with minimal tissue inflammation (Andermann et al., 2013). The current study was initially done on anesthetized animals to examine and validate the new approach, which later revealed similar organizational differences between LC and DC in awake animals. Although imaging through a microprism and 1-mm cover glass appears to have minimal impact on image quality compared to direct 2P imaging (Chia & Levene, 2009), it was important to compare LC imaging via microprism with the direct 2P imaging of the dorsal surface to validate the new imaging method. We found no significant differences between LC response properties when imaging via microprism or imaging directly from the dorsal surface. However, we did observe significant functional differences between the LC and the DC.

Responsiveness to spectral and temporal features

While the LC showed inconsistent and fragmented tonotopy, the DC showed a consistent one, as previously reported (Barnstedt et al., 2015; A. B. Wong & Borst, 2019). However, the surface of the DC was biased to lower frequencies compared to LC, consistent with previous imaging and recording studies (Ito et al., 2014; Romand & Ehret, 1990; Shen et al., 2003; Stiebler & Ehret, 1985). Despite their response to pure tones, the DC and LC, particularly in the matrix, were more responsive to AM-noise, which fits their function as nonlemniscal divisions (Chen, Cheng, Ito, & Song, 2018; Faye-Lund & Osen, 1985; Loftus et al., 2008).

The spectral and temporal features of animal vocalization and speech are not independent (Singh & Theunissen, 2003). We observed that LC neurons, particularly in the matrix, had greater responsiveness to spectral and temporal modulation than the DC. Given the role of inhibition in the spectral and temporal processing (Cai et al., 2018; Chang, Bao, Imaizumi, Schreiner, & Merzenich, 2005; Wehr & Zador, 2003), GABAergic cells of the modules could play an important role to increase the spectral and temporal modulation indices, especially at lower sound levels, which could be critical for localizing objects in noisy environments given the importance of spectral and temporal cues for sound localization (Davis, Ramachandran, & May, 2003; Goodman, Benichoux, & Brette, 2013; Sinex, 2005). Therefore, future work is critical to image the functional responses of the GABAergic cells of the modules and compare them with the non-GABAergic cells inside and outside the modules.

Somatosensory vs. auditory responses

In the LC, we observed that most auditory-responsive neurons, particularly onset-responsive neurons, were found in the matrix, which receives mostly auditory inputs (A. M. Lesicko et al., 2016 [DOI](#)). However, a subpopulation of acoustically-driven offset-responsive neurons preferably resides in the modules compared to the matrix. Given that the offset response could be generated as a rebound to the termination of the ongoing inhibition during the sound presentation (Xu, 2014 [DOI](#)), GABAergic cells in the modules could play an important role in the coding of sound termination as found before in the cortex (Solyga & Barkat, 2021 [DOI](#)). Because some of the offset acoustically-driven cells inside the modules were also activated by the somatosensory stimulation, this fraction of cells could then encode the termination of somatosensory-associated sounds through the local inhibitory circuits of the modules. Under in-vivo recording of the mouse IC, the offset responses were previously reported from cells that were non-GABAergic and located in the caudal and shallower parts of the IC (Kasai, Ono, & Ohmori, 2012 [DOI](#)).

Consistent with previous reports (Aitkin et al., 1978 [DOI](#); Aitkin, Kenyon, & Philpott, 1981 [DOI](#); Jain & Shore, 2006 [DOI](#)), we observed that combined auditory-somatosensory stimulation generally suppressed neural responses to auditory stimuli and that this suppression was most prominent in the LC matrix. Mechanistically, it is not known if such suppressive responses are created in the LC or are inherited from other brain regions since auditory-somatosensory interactions occur as early in the cochlear nucleus, and are also found in the IC, thalamus, and cortex (Aitkin et al., 1981 [DOI](#); Haenggeli, Pongstaporn, Doucet, & Ryugo, 2005 [DOI](#); Kimura, Imbe, Donishi, & Tamai, 2007 [DOI](#); Li & Mizuno, 1997 [DOI](#); Lohse, Dahmen, Bajo, & King, 2021 [DOI](#); Shore, 2005 [DOI](#); Shore, Vass, Wys, & Altschuler, 2000 [DOI](#); C. Wong, Chabot, Kok, & Lomber, 2015 [DOI](#)), reviewed in (Brunelle & Llano, 2023 [DOI](#); Lohse et al., 2022 [DOI](#)). We note that a systematic exploration of the stimulus space was not conducted here such that other acoustic or somatosensory stimuli, different relative amplitudes or timings could have produced enhancement rather than suppression. Future work will be needed to further explore this stimulus space to more fully understand multisensory processing in the LC.

Conclusions

Here, using a novel microprism approach to image the LC in anesthetized and awake mice, we observed that 1) DC displayed tonotopic organization which was not consistently present in the LC, 2) DC neurons were generally more sound-responsive than LC neurons but that neurons in both regions had preferential responses to spectrally-and temporally-complex stimuli, 3) that matrix regions of the LC were more responsive to spectrally-and temporally-complex stimuli than modules and that modules were more responsive to somatosensory stimuli and 4) that bimodal stimulation led to suppressive responses that were more prominent in the matrix and enhancement that was more prominent in modules. These data highlight significant organizational differences between the LC and DC and between modules and matrix in the LC. Future work using microprisms in the LC will help to reveal the potential differential role of matrix vs. module neurons in multisensory integration tasks in behaving animals.

Materials and methods

Animal subjects

Male mice of 8-12 weeks of age were used. Males were used as our preliminary data suggest male/female differences in DC organization (Ibrahim, Shinagawa, Xiao, Asilador, & Llano, 2022 [DOI](#)) and will be published separately. GAD67-GFP knock-in mice (developed and shared with permission from Dr. Yuchio Yanagawa at Gunma University and obtained from Dr. Douglas Oliver

at the University of Connecticut), where GFP is exclusively expressed in GABAergic cells (Tamamaki et al., 2003 [\[4\]](#)), were used to visualize the GABAergic cells and LC modules. To simultaneously monitor calcium signals and visualize the GFP+ cells, Tg(Thy1-jRGECO1a)GP8.20Dkim/J mice (Jackson Laboratory, Stock# 030525) were crossed with GAD67-GFP knock-in mice to generate GAD67-GFPxThy1-jRGECO1a hybrid mice. The mice were housed and bred at the animal research facility of The Beckman Institute for Advanced Science and Technology. The animal care facilities are approved by the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC). To phenotype the GFP-positive animals, a single-photon fluorescence microscope was used for the transcranial examination of the pups at postnatal day 4 using excitation: 472/30 nm and emission 520/35 nm filters along with dichroic 505 nm long pass at low power magnification (2.5x/0.08, Olympus objective, MPlanFL N, Japan). The positive pups exhibited a green fluorescence in the cerebellum, the cerebral cortex, and the olfactory bulb and were only kept with their parents until weaning. The genotyping for jRGECO1a was done using a probe composed of a forward primer: GCCGCCGAGGTCAAGA & a reverse primer: TCCAACCTTGATGTCGACGATGTAG by Transnetyx™ (Transnetyx, USA). Samples of the mice's tail snip around one month of age were used for genotyping. All applicable guidelines for the care and use of animals were followed. All procedures were approved by the Institutional Animal Care and Use Committee (IACUC).

Surgery

The general procedures of craniotomy were previously described (Goldey et al., 2014 [\[4\]](#)), with modifications to place craniotomy over the IC. Before surgery, mice were anesthetized with a mixture of ketamine, xylazine, and acepromazine (100 mg/kg, 3 mg/kg, and 3mg/kg, respectively) delivered intraperitoneally. The anesthesia was maintained during the surgery and imaging using only ketamine (100 mg/kg). To prevent neural edema during or after the craniotomy, an intramuscular injection of dexamethasone sodium (4.8 mg/kg) was given just before the surgery using an insulin syringe. After placing the animal in the stereotaxic apparatus (David Kopf Instruments, USA), both eyes were protected by applying Optixcare lubricant eye gel (Aventix Animal Health, Canada). The hair on the scalp was then removed by massaging the scalp with a depilatory cream (Nair) using a cotton-tipped applicator and leaving the cream on the scalp for 4-5 minutes. The cream was then removed by a thin plastic sheet (flexible ruler) to leave a hair-free area on the scalp. The remaining tiny hairs were then removed by alcohol swab and the area was then sterilized by applying 10% Povidone Iodine (Dynarex, USA) using a sterile cotton-tipped applicator. The medial incision was made with a scalpel blade #10, and 0.2 ml of 0.5% lidocaine was injected intradermally into the scalp. The skin extending from the medial line to the temporalis muscle was completely removed using a pair of microscissors to produce a wide skinless area above the skull. Number 5/45 forceps were used to remove any remaining periosteum. The remaining dried or firmly attached pieces of periosteum were removed with a scalpel blade #10. The skull was cleaned with sterile saline and dried with gently pressurized air. Using the stereotaxic apparatus, a wide area of ~3 x 4 mm above the left IC was made. A micro drill bit (size #80, Grainger, USA) was used to drill through the skull starting from the rostralateral (**Figure 1A** [\[4\]](#), yellow circle) region to lambda (**Figure 1A** [\[4\]](#), black circle) following the border in a clockwise direction (**Figure 1A** [\[4\]](#), dotted black rectangle). To prevent overheating of the superficial brain tissues and to mitigate the occasional spurts of skull bleeding during the drilling, ice-cold sterile saline was used to intermittently irrigate the surface. A stream of pressurized air was also applied during the drilling procedures to prevent overheating and remove the debris produced by the drilling. Caution was taken not to pierce the dura when performing the craniotomy while crossing the sagittal or the lambdoid sutures to avoid damaging the underlying sinuses. After drilling, the skull was irrigated in sterile saline and the bone flap (the undrilled bone over the craniotomy area) was gently examined for complete separation from the rest of the skull (**Figure 1B** [\[4\]](#)). Using a pair of no. 5/45 forceps, the bone flap was gently removed. To control the bleeding if it occurred, a piece of sterile hemostatic gel (Adsorbable Gelatin Sponge USP, Haemosponge, GBI, India) which was pre-soaked in ice-cold saline, was applied to the bleeding spot. Once the bleeding ceased, the brain was kept covered in sterile saline. In some surgeries, the

dura was peeled off from the surface of the IC while removing the bone flap. In the surgeries where the dura remained intact, a Bonn microprobe (F.S.T, Germany, Item # 10032-13) was used to pierce the dura in an area that is not above the IC and devoid of cortical vasculature (e.g., a part of exposed cerebellum). After piercing, the dura was carefully and gently lifted, and a pair of no. 5/45 forceps were used to grab the dura to gently tear it to the extent of the transverse sinus to avoid bleeding of the major venous structures.

Using the stereotaxic apparatus, a 30° diamond knife (F.S.T, Germany, Item # 10100-30) was placed in its vertical position. From the medial line, the knife was placed at the lateral horizon of the IC. The knife was then inserted 1.5 mm ventrally. Caution was taken to avoid hitting any major blood vessels to avoid bleeding. From its position, the knife was retracted 0.3-0.5 mm medially to create an initial pocket for the insertion of the microprism (**Figures 1C-E** [↗](#)). The microprism (1.5 mm-silver coated, Tower Optical Corporation) was then inserted, so one of its sides was parallel to the lateral surface of the diamond blade, the other side was facing up, and its hypotenuse was facing ventrally (**Figures 1F and G** [↗](#)). The wooden piece of a cotton swab was carefully driven down to push the microprism until its top surface was at the same level as the IC dorsal surface. Keeping the microprism in its place with the wooden end of a swab, the diamond knife was carefully retracted upward leaving the microprism behind (**Figure 1H** [↗](#)). If the retraction of the diamond knife pulled the prism up, the wooden piece would be used to push it down simultaneously with the knife movement. To control the bleeding, a piece of sterile hemostatic gel was used as before. The tip of the diamond knife was used to secure the microprism in its position for at least 10 minutes (**Figure J**). During that time, the solution of the 1% agarose was prepared for use. A pair of forceps were then used to gently place a sterile 5 mm cover glass #1 (Thomas Scientific, USA) over the skull covering the opened area above the IC. The cover glass was secured by a wooden trimmed piece of sterile cotton swab by gently pressing the cover glass from the top. Since the surface of the IC was located below the surfaces of both the cerebellum and cerebral cortex, the cover glass usually traps about a 0.7-1.0 mm thick sheet of sterile saline. The movement of the liquid medium caused by heart pulsation was found to be a source of motion artifact during imaging, so the gap between the surface of the IC and the cover glass was filled with 1% agarose gel that was made in saline at a temperature range of 33-35°C (**Figure 1K** [↗](#)). The extra agarose gel pieces were cut off and removed by the scalpel blade and the wet skull was then dried out using pressurized air. A titanium headpost as described before ([Goldey et al., 2014](#) [↗](#)) was glued carefully on the top of the skull to be at the same level as the cover glass. Following the manufacturer's instructions, the C&B Metabond (Parkell, Japan) was used to secure the headpost in its place. Excluding the steps of the microprism insertion, the same procedures were followed for DC imaging from the surface (**Figure 1L** [↗](#)). For awake preparations, the same procedures were followed under isoflurane anesthesia starting with 4% of isoflurane as an induction dose with 1-2% isoflurane during the surgery. The animal was transferred under the microscope objective and left to recover from isoflurane anesthesia for 1 hour before imaging. To ensure the animal's recovery, the animal was visually inspected for its breathing rate ([Ewald, Werb, & Egeblad, 2011](#) [↗](#)) and for ongoing limb movements.

Acoustic stimulation

Using a custom-made MATLAB (The MathWorks, Natick, MA, USA) code, either 500 ms pure tone or AM-noise was generated. Thirty-five (5×7) pure tones with a series of sound pressure levels (40, 50, 60, 70, 80 dB SPL) and a series of carrier frequencies (5000-40000 Hz with a half-octave gap) were presented with a cosine window. Forty-five (5×9) 100% AM white noise bursts with a series of sound pressure levels (40, 50, 60, 70, 80 dB SPL) and modulation frequencies (0, 2, 4, 8, 16, 32, 64, 128, 256 Hz) were generated. The stimuli of either pure tone or AM-noise combinations were played in random sequence to the mice with a 600 ms interstimulus interval (ISI) by a TDT RP2.1 processor (Tucker-Davis Technologies, US) and delivered by a TDT ES1 speaker (Tucker-Davis Technologies, US). In other sets of experiments, 500 ms of nonmodulated broadband noise was presented 5 times with ISI of 600 ms at 80 dB SPL to examine the acoustic activity between modules and matrix compared to somatosensory stimulation.

The output of the TDT ES1 speaker was calibrated using a PCB 377A06 microphone, which feeds a SigCal tool to generate a calibration file for all tested frequencies (5–40 kHz). To enable the custom-made MATLAB code to read this calibration file, the values were first processed by MATLAB signal processing toolbox (sptool) to generate a 256-tap FIR filter to apply the calibration using the following parameters [arbitrary magnitudes, least square, order: 256, sampling rate: 97,656.25, frequency vector (5–40 kHz), amplitude vector (40–80 dB SPL), and weight vector ones (1128)].

Somatosensory stimulation

The somatosensory stimulation was done by deflecting the right whiskers of the animals at an average excursion of 1.2 mm with a 50 Hz rate over 500 ms to match the acoustic stimulus. A total of five stimuli were presented with 600 ms ISI. The whisker deflection was performed using a Brüel & Kjær vibrator (V203 10/32 UNF-CE, United Kingdom) that was driven by a wave generator (4063, BK precision, USA) after amplifying the signals with an SLA1 100 W power amplifier (ART, USA).

Two-Photon imaging

Immediately after surgery, the anesthetized animal was taken and secured under the microscope objective by clamping the arms of the head post to two perpendicular metal posts mounted on the microscope stage. A custom-built 2P microscope was used. The optical and the controlling components were supplied from Bruker, Olympus, and Thorlabs. The imaging of the DC was made using a 20x water-immersion objective (LUMPlanFI/IR, 20X, NA: 0.95, WD: 2 mm; Olympus Corporation, Tokyo, Japan), while the lateral surface of the IC was imaged by the long working distance 16x water-immersion objective (N16XLWD-PF - CFI LWD Plan Fluorite Objective, 16X, NA: 0.8, WD: 3.0 mm; Nikon, Tokyo, Japan) to be able to reach the focal point of the microprism at its hypotenuse. For imaging both the GFP or jRGECO1a signals, the excitation light was generated by InSight X3 laser (Spectra-Physics Lasers, Mountain View, CA, USA) tuned to a wavelength of 920 or 1040 nm, respectively. A layer of a 1:1 mixture of wavelength multipurpose ultrasound gel (National Therapy, Canada) with double deionized water was used to immerse the objective. This gel was able to trap the water and reduce its evaporation during imaging. The emitted signals were detected by a photomultiplier tube (Hamamatsu H7422PA-4, Japan) following a t565lp dichroic and a Chroma barrier et525/70m filter for GFP and et595/50m filter for jRGECO1a signals. Images (512×512 pixels) were collected at a frame rate of 29.9 Hz in the resonant galvo mode. Data were collected from the dorsal surface of the IC by scanning the surface of the IC based on the GFP and jRGECO1a signals through the medial and lateral horizons of the IC. Generally, the scanning was started by moving the 20x objective to the most ventromedial position, where there is no ability to see any cells (**Figure 2B** [↗](#), medial horizon). Then, the objective was started to slowly move dorsolaterally to get cells in focus (**Figure 1M** [↗](#)), where the average depth of the first field of view was 169 μm.

The lateral movement of the objective was intended to be aligned and parallel to the curvature of the DC surface indicated by getting the superficial cells in focus to avoid any possibility of imaging the upper layers of the central nucleus of the IC, so the average depth of the most superficial point of the imaging was 92 μm. The scanning was ended by moving the objective to the most ventrolateral position of the IC, where there is no ability to see GFP laterally (**Figure 2B** [↗](#), lateral horizon), where the last field of view was collected at an average depth of 225 μm.

Generally, each field of view was selected based on the expression of GFP signals as an indicator for the cells in focus and being acoustically active by using a search stimulus that was 500 ms broadband noise with zero modulation at 80 dB SPL. The z-axis of the objective was determined relative to the pia surface. The relative position of each region was tracked through the micromanipulator (MP-285, Sutter Instruments) that controlled the microscope objective.

The positions of imaged areas were further aligned across animals to a common coordinate using the midline, lateral extremes, and the vasculature landmarks taken via imaging of the GFP signals of the cranial window with low magnification (4x Olympus objective, 4×4 binning, 50 ms frame time) using a Lumen 200 bulb (Prior Scientific Inc, USA) and CoolSNAP MYO camera (Photometrics, USA, excitation: 488 nm and emission: 515–550 nm). Imaging of the lateral surface of the IC via microprism was done similarly. In both cases, the frame timing of the scanner and the sound stimuli were both digitized and time-locked using a Digidata 1440A (Molecular Devices, Sunnyvale, CA, USA) with Clampex v. 10.3 (Molecular Devices, Sunnyvale, CA, USA).

Laser lesion

The laser lesion was produced using a 720 or 820 nm laser beam at 74–80 mW power at the level of the objective for 100 to 250 seconds depending on the size of the lesioned area, the speed of laser scanning (frame rate), and the depth of the lesion (z-axis length). In brief, the area of interest and the dimension of the laser lesion were initially determined using 920 nm as indicated by the GFP signals through scanning one of the GABAergic modules. The depth of the GABAergic modules was then determined by counting the optical slices starting from their superficial surface. The tunable laser was then switched from 920 nm to the shorter wavelength (720 or 820 nm) to scan the same area and depth at a rate (1–2 frames/ second).

Data processing

Data collection

The data were collected as separate movies (512×512 pixels) for each pure tone or AM-noise runs in a resonant galvo mode. Depending on the amplitude and frequency combinations for each type of acoustic stimulus, 40- or 52-second periods were assigned as a movie's length for pure tone (35 stimulus combinations), or AM-noise (45 stimulus combinations), respectively. Using ImageJ software, the z-projection was used to compute one single image representing either the sum, the standard deviation, or the median of all the image sequences in the movie. Based on these single images, the region of interest (ROI) was manually drawn around each detectable cell body. **Figures 3A and B** represent the sum of the images sequences of the movie collected from the dorsal surface, or the lateral surface of the IC via microprism, respectively. For the subsequent processing steps, Python open-source libraries were used (listed below).

Motion Correction and Filtering

The imread function from the OpenCV library was used in grayscale mode to import images into a numpy array from a single folder containing TIFF images. The array was then exported as a single TIFF stack into a temporary folder using the mimwrite function from the imageio library, and the process was repeated for each folder. The NoRMCorre algorithm (Pnevmatikakis & Giovannucci, 2017) embedded in the CaImAn library (Giovannucci et al., 2019) was used to apply motion correction to each of the TIFF files. The data were then imported into a numpy array, rounded, and converted to 16-bit integers. The images were filtered using a 2D Gaussian filter with a sigma value of 1 (surface View) / 2 (prism View) in each direction, then a 1D Gaussian temporal filter with a sigma value of 2 was applied using the ndimage.gaussian_filter and ndimage.gaussian_filter1d function from the scipy library, respectively.

Data Extraction

The ROI sets, which were manually created using ImageJ, were imported using the read_roi_zip function from the read_roi library. The sets were then used to create two masks; one mask was used as a replica of the ROIs and the second mask was made around the original ROI (roughly four

times larger in area). The smaller mask was applied to find the average pixel value within each ROI, while the larger mask was applied to find the average pixel value of the neuropil. The neuropil correction was applied using the following equation (Akerboom et al., 2012 [DOI](#));

$$\text{Corrected value} = \text{Date value} - (0.4 \times \text{Neuropil value})$$

To identify the calcium signals, $\Delta f/f$ was calculated by using the following equation.

$$\frac{\Delta f}{f} = \frac{(\text{Date value} - \text{Background value})}{\text{Background value}}$$

where the background value is the slope estimating the background levels with fluorescence bleaching factored in. The data were then reorganized so that all segments with the same stimulus frequency and stimulus amplitude were grouped. The area under the curve (AUC) of the calcium signals of the excitatory responses only was then used as a metric to determine the magnitude of the response and for subsequent analysis (Berens et al., 2018 [DOI](#); Overk, Rockenstein, Florio, Cheng, & Masliah, 2015 [DOI](#)).

Cell Flagging

The correlation coefficient between each of the trials was calculated using the stats. Pearson function from the scipy library. The average correlation coefficient was calculated for each stimulus frequency and stimulus amplitude. Similar to previous work (A. B. Wong & Borst, 2019 [DOI](#)), if the average correlation coefficient was above the threshold of 0.6, the cell was flagged as being responsive to that stimulus combination of frequency and amplitude, and the best tone and modulation frequencies were calculated for every cell (Barnstedt et al., 2015 [DOI](#)). Knowing the best tonal frequency (BTF) and modulation frequency (BMF) for each neuron enabled us to calculate the SMI and TMI at each sound level following these equations;

$$\text{SMI} = \frac{\text{Response to BTF} - \text{Response to unmodulated noise}}{\text{Response to BTF} + \text{Response to unmodulated noise}}$$

The SMI has values ranging from (-1 to 1). While neurons with values closer to 1 are more responsive to pure tone, neurons with values closer to -1 are more responsive to noise with no modulation.

$$\text{TMI} = \frac{\text{Response to BMF} - \text{Response to unmodulated noise}}{\text{Response of BMF} + \text{Response to unmodulated noise}}$$

Similarly, the TMI has values ranging from (-1 to 1). While neurons with values closer to 1 are more responsive to AM-noise, neurons with values closer to -1 are more responsive to noise with no modulation.

For the comparison between somatosensory and auditory responses, the threshold of 0.4 was used as a cutoff threshold to determine the responsive cells to each stimulation. The average response of each somatosensory, acoustic (500 ms of unmodulated noise), or simultaneous somatosensory and acoustic (bimodal) stimulation was then calculated for each responsive cell.

To examine the effect of somatosensory stimulation on auditory responses, the response index (RI) was calculated for the auditory-responsive cells using the following equation;

$$\text{RI} = \frac{\text{Auditory response to bimodal stimulation}}{\text{Auditory response to auditory stimulation only}}$$

Neurons with RI values > 1 have enhanced or higher responses induced by simultaneous auditory and somatosensory stimulation, while neurons with RI values < 1 have downregulated or suppressed auditory responses induced by auditory and simultaneous somatosensory stimulation.

Map Generation

The average radius for all ROIs was calculated to ensure that all cells on the tonotopic map had uniform radii. A color key was also generated for every corresponding map.

A blank canvas was generated using the Image module from the pillow library, and a circle was created for each cell and filled with a corresponding color to its value.

Statistics

All statistical analyses and graphs were made using Origin Pro 2022 software. The normality of the distributions of the data was examined using a Kolmogorov-Smirnov test. The Mann-Whitney test as well as two-sample or paired t-test were used to examine the difference in data distribution or the difference between samples of non-normal and normal distribution, respectively. Either one-way or two-way ANOVA as well as Fisher as a post hoc test were used for the mean comparison between different groups of normal distribution. The Chi-square (χ^2) test was used to compare the samples based on cell numbers of their cellular population. The cumulative frequency function was computed to calculate the fraction of cells tuned at each frequency based on their best frequency. Differences were deemed significant when p-value < 0.05. The upper and lower borders of the each box plot are at the 25th and 75th percentile, with whiskers at 95% spread.

Histology

After the 2P imaging, the mouse was transcardially perfused with 4% paraformaldehyde (PFA) solution in PBS solution. The isolated brain was then placed overnight in PFA for a complete fixation. The brain was then moved from the PFA to a graded sucrose solution (10, 20, and 30%) until it sunk. 50 μ m coronal sections were taken at the level of the IC using a Leica cryostat. The sections were then mounted on a slide, coverslipped, and imaged using an confocal microscope for imaging the GFP signals as an indicator for the GABAergic cells in the GAD67-GFP knock-in mouse (excitation: 488 nm and emissions: 515–550 nm).

Artwork

The cartoon of the mouse brain in **Figures 1** [C](#), [F](#), and [I](#) was obtained from Brain Explorer 3D viewer Allen Mouse Brain (<https://connectivity.brain-map.org/3d-viewer?v=1> [C](#)) and modified to be suitable for illustration. All figures were designed and made using Adobe Illustrator (Adobe, San Jose, CA). To keep working within the Adobe environment to avoid losing the resolution of the figures, Adobe Photoshop (Adobe, San Jose, CA) was used to crop the borders of some images to save space.

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Competing interests

On behalf of all authors, there are no financial or non-financial competing interests to be declared.

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Reviewer #1 (Public Review):

Summary:

In this paper, the authors provide a characterisation of auditory responses (tones, noise, and amplitude-modulated sounds) and bimodal (somatosensory-auditory) responses and interactions in the higher-order lateral cortex (LC) of the inferior colliculus (IC) and compare these characteristics with the higher order dorsal cortex (DC) of the IC - in awake and anaesthetised mice. Dan Llano's group has previously identified gaba'ergic patches (modules) in the LC distinctly receiving inputs from somatosensory structures, surrounded by matrix regions receiving inputs from the auditory cortex. They here use 2P calcium imaging combined with an implanted prism to - for the first time - get functional optical access to these subregions (modules and matrix) in the lateral cortex of IC in vivo, in order to also characterise the functional difference in these subparts of LC. They find that both DC and LC of both awake and anaesthetised mice appear to be more responsive to more complex sounds (amplitude-modulated noise) compared to pure tones and that under anesthesia the matrix of LC is more modulated by specific frequency and temporal content compared to the gabaergic modules in LC. However, while both LC and DC appear to have low-frequency preferences, this preference for low frequencies is more pronounced in DC. Furthermore, in both awake and anesthetized mice, somatosensory inputs are capable of driving responses on their own in the modules of LC, but very little (possibly not at all) in the matrix. However, bimodal interactions may be different under awake and anesthesia in LC, which warrants deeper investigation by the authors: They find, under anesthesia, more bimodal enhancement in modules of LC compared to the matrix of LC and bimodal suppression dominating the matrix of LC. In contrast, under awake conditions bimodal enhancement is almost exclusively found in the matrix of LC, and bimodal suppression dominates both matrix and modules of LC.

The paper provides new information about how subregions with different inputs and neurochemical profiles in the higher-order auditory midbrain process auditory and multisensory information, and is useful for the auditory and multisensory circuits neuroscience community.

Strengths:

The major strength of this study is undoubtedly the fact that the authors for the first time provide optical access to a subcortical region (the lateral cortex of the inferior colliculus (i.e. higher order auditory midbrain)) which we know (from previous work by the same group) have optically identifiable subdivisions with unique inputs and neurotransmitter release, and plays a central role in auditory and multisensory processing. A description of basic auditory and multisensory properties of this structure is therefore very useful for understanding auditory processing and multisensory interactions in subcortical circuits.

Weaknesses:

I have divided my comments about weaknesses and improvements into major and minor comments. All of which I believe are addressable by the reviewers to provide a more clear picture of their characterisation of the higher-order auditory midbrain.

Major comment:

1. The differences between multisensory interactions in LC in anaesthetised and awake preparations appear to be qualitatively different, though the authors claim they are similar (see also minor comment related to figure 10H for further explanation of what I mean). However, the findings in awake and anaesthetised conditions are summarised differently, and plotting of similar findings in the awake figures and anaesthetised figures are different - and different statistics are used for the same comparisons. This makes it very difficult to assess how multisensory integration in LC is different under awake and anaesthetised conditions. I suggest that the authors plot (and test with similar statistics) the summary plots in Figure 8 (i.e. Figure 8H-K) for awake data in Figure 10, and also make similar plots to Figures 10G-H for anaesthetised data. This will help the readers understand the differences between bimodal stimulation effects on awake and anaesthetised preparations - which in its current form, looks very distinct. In general, it is unclear to me why the awake data related to Figures 9 and 10 is presented in a different way for similar comparisons. Please streamline the presentation of results for anaesthetised and awake results to aid the comparison of results in different states, and explicitly state and discuss differences under awake and anaesthetised conditions.

2. The claim about the degree of tonotopy in LC and DC should be aided by summary statistics to understand the degree to which tonotopy is actually present. For example, the authors could demonstrate that it is not possible/or is possible to predict above chance a cell's BF based on the group of other cells in the area. This will help understand to what degree the tonotopy is topographic vs salt and pepper. Also, it would be good to know if the gaba'ergic modules have a higher propensity of particular BFs or tonotopic structure compared to matrix regions in LC, and also if general tuning properties (e.g. tuning width) are different from the matrix cells and the ones in DC.

3. Throughout the paper more information needs to be given about the number of cells, sessions, and animals used in each panel, and what level was used as n in the statistical tests. For example, in Figure 4 I can't tell if the 4 mice shown for LC imaging are the only 4 mice imaged, and used in the Figure 4E summary or if these are just examples. In general, throughout the paper, it is currently not possible to assess how many cells, sessions, and animals the data shown comes from.

4. Throughout the paper, to better understand the summary maps and plots, it would be helpful to see example responses of the different components investigated. For example, given that module cells appear to have more auditory offset responses, it would be helpful to see what the bimodal, sound-only, and somatosensory responses look like in example cells in LC modules. This also goes for just general examples of what the responses to auditory and somatosensory inputs look like in DC vs LC. In general example plots of what the responses actually look like are needed to better understand what is being summarised.

Reviewer #2 (Public Review):

Summary:

The study describes differences in responses to sounds and whisker deflections as well as combinations of these stimuli in different neurochemically defined subsections of the lateral and dorsal cortex of the inferior colliculus in anesthetised and awake mice.

Strengths:

The main achievement of the work lies in obtaining the data in the first place as this required establishing and refining a challenging surgical procedure to insert a prism that enabled the authors to visualise the lateral surface of the inferior colliculus. Using this approach, the authors were then able to provide the first functional comparison of neural responses inside and outside of the GABA-rich modules of the lateral cortex. The strongest and most

interesting aspects of the results, in my opinion, concern the interactions of auditory and somatosensory stimulation. For instance, the authors find that a) somatosensory-responses are strongest inside the modules and b) somatosensory-auditory suppression is stronger in the matrix than in the modules. This suggests that, while somatosensory inputs preferentially target the GABA-rich modules, they do not exclusively target GABAergic neurons within the modules (given that the authors record exclusively from excitatory neurons we wouldn't expect to see somatosensory responses if they targeted exclusively GABAergic neurons), and that the GABAergic neurons of the modules (consistent with previous work) preferentially impact neurons outside the modules, i.e. via long-range connections.

Weaknesses:

While the findings are of interest to the subfield they have only rather limited implications beyond it. The writing is not as precise as it could be. Consequently, the manuscript is unclear in some places. For instance, the text is somewhat confusing as to whether there is a difference in the pattern (modules vs matrix) of somatosensory-auditory suppression between anesthetized and awake animals. Furthermore, there are aspects of the results which are potentially very interesting but have not been explored. For example, there is a remarkable degree of clustering of response properties evident in many of the maps included in the paper. Taking Figure 7 for instance, rather than a salt and pepper organization we can see auditory responsive neurons clumped together and non-responsive neurons clumped together and in the panels below we can see off-responsive neurons forming clusters (although it is not easy to make out the magenta dots against the black background). This degree of clustering seems much stronger than expected and deserves further attention.

Reviewer #3 (Public Review):

The lateral cortex of the inferior colliculus (LC) is a region of the auditory midbrain noted for receiving both auditory and somatosensory input. Anatomical studies have established that somatosensory input primarily impinges on "modular" regions of the LC, which are characterized by high densities of GABAergic neurons, while auditory input is more prominent in the "matrix" regions that surround the modules. However, how auditory and somatosensory stimuli shape activity, both individually and when combined, in the modular and matrix regions of the LC has remained unknown.

The major obstacle to progress has been the location of the LC on the lateral edge of the inferior colliculus where it cannot be accessed *in vivo* using conventional imaging approaches. The authors overcame this obstacle by developing methods to implant a microprism adjacent to the LC. By redirecting light from the lateral surface of the LC to the dorsal surface of the microprism, the microprism enabled two-photon imaging of the LC via a dorsal approach in anesthetized and awake mice. Then, by crossing GAD-67-GFP mice with Thy1-jRGECO1a mice, the authors showed that they could identify LC modules *in vivo* using GFP fluorescence while assessing neural responses to auditory, somatosensory, and multimodal stimuli using Ca²⁺ imaging. Critically, the authors also validated the accuracy of the microprism technique by directly comparing results obtained with a microprism to data collected using conventional imaging of the dorsal-most LC modules, which are directly visible on the dorsal IC surface, finding good correlations between the approaches.

Through this innovative combination of techniques, the authors found that matrix neurons were more sensitive to auditory stimuli than modular neurons, modular neurons were more sensitive to somatosensory stimuli than matrix neurons, and bimodal, auditory-somatosensory stimuli were more likely to suppress activity in matrix neurons and enhance activity in modular neurons. Interestingly, despite their higher sensitivity to somatosensory stimuli than matrix neurons, modular neurons in the anesthetized prep were far more responsive to auditory stimuli than somatosensory stimuli (albeit with a tendency to have offset responses to sounds). This suggests that modular neurons should not be thought of as primarily representing somatosensory input, but rather as being more prone to having their

auditory responses modified by somatosensory input. However, this trend was reversed in the awake prep, where modular neurons became more responsive to somatosensory stimuli than auditory stimuli. Thus, to this reviewer, the most intriguing result of the present study is the dramatic extent to which neural responses in the LC changed in the awake preparation. While this is not entirely unexpected, the magnitude and stimulus specificity of the changes caused by anesthesia highlight the extent to which higher-level sensory processing is affected by anesthesia and strongly suggest that future studies of LC function should be conducted in awake animals.

Together, the results of this study expand our understanding of the functional roles of matrix and module neurons by showing that responses in LC subregions are more complicated than might have been expected based on anatomy alone. The development of the microprism technique for imaging the LC will be a boon to the field, finally enabling much-needed studies of LC function in vivo. The experiments were well-designed and well-controlled, and the limitations of two-photon imaging for tracking neural activity are acknowledged. Appropriate statistical tests were used. There are three main issues the authors should address, but otherwise, this study represents an important advance in the field.

1. Please address whether the Thy1 mouse evenly expresses jRGECO1a in all LC neurons. It is known that these mice express jRGECO1a in subsets of neurons in the cerebral cortex, and similar biases in the LC could have biased the results here.
2. I suggest adding a paragraph or two to the discussion to address the large differences observed between the anesthetized and awake preparations. For example, somatosensory responses in the modules increased dramatically from 14.4% in the anesthetized prep to 63.6% in the awake prep. At the same time, auditory responses decreased from 52.1% to 22%. (Numbers for anesthetized prep include auditory responses and somatosensory + auditory responses.). In addition, the tonotopy of the DC shifted in the awake condition. These are intriguing changes that are not entirely expected from the switch to an awake prep and therefore warrant discussion.
3. For somatosensory stimuli, the authors used whisker deflection, but based on the anatomy, this is presumably not the only somatosensory stimulus that affects LC. The authors could help readers place the present results in a broader context by discussing how other somatosensory stimuli might come into play. For example, might a larger percentage of modular neurons be activated by somatosensory stimuli if more diverse stimuli were used?